

Project Submission Pro-Forma

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- ☒ A glossary of terms (where appropriate)
- ☒ A clear statement of my project objectives
- ☒ A full reference list (the Harvard referencing style is recommended for WMG)
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Design and Evaluation of an Evidence-First Multi-Agent Workflow for UK Early-Stage Portfolio Reporting

by

Faye Niu

Dissertation submitted in partial fulfilment for the Degree of Master of Science in
Applied Artificial Intelligence

WMG
University of Warwick

Submitted _____

Declaration

I have read and understood the rules on cheating, plagiarism and appropriate referencing as outlined in my handbook and I declare that the work contained in this assignment is my own, unless otherwise acknowledged.

No substantial part of the work submitted here has also been submitted by me in other assessments for this or previous degree courses, and I acknowledge that if this has been done an appropriate reduction in the mark I might otherwise have received will be made.

I acknowledge the use of OpenAI Codex (<https://openai.com/codex/>) as a generative artificial-intelligence tool. It was used only to support report structure, visualisations, research and source-checking, and feedback on the writing. It was not used to implement or validate the prototype. I reviewed the proposed outputs against the repository evidence and cited sources, modified or rejected them where necessary, and remain responsible for the submitted work. Appendix J records the prompts, affected work and the way outputs were checked and changed.

Project definition for my degree (as copied from the Specific Course Regulations for the course)

This substantial individual project is designed to strengthen your academic and research skills, while allowing you to explore a specific area of artificial intelligence that reflects your interests and career goals. The project will be presented as a written dissertation, submitted at the end of the academic year. You are expected to devote an appropriate portion of your time and intellectual effort to the project during the course i.e. approximately 600 hours over the course of the academic year.
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My project relates to this definition in the following way

My project is an individual artificial intelligence project about creating public-company research with clear supporting evidence. I designed and built a local prototype for one user and prepared a study plan. The workflow starts with a Companies House number, which a named person must check. It saves public sources, exact passages, citations and time rules before it produces claims for review. OpenAI can help find possible public URLs, but the system then collects and checks each source through its own protected process. I have not claimed a completed live company-research run or a comparison with people. The workflow keeps extraction separate from verification, records missing, conflicting and unsupported evidence, and requires a named person to approve the current version before export. The dissertation reports the prototype, study plan, checked engineering evidence, limitations and remaining work. This project therefore supports my academic and research skills in an area of artificial intelligence that matches my interests and career goals.

Abstract

Early-stage portfolio reporting combines company submissions with public evidence that may be incomplete, delayed or inconsistent. The current process has three broad stages: company information is collected, aggregated into an executive summary, and prepared for review before use. The main risks are incorrect company identity, confusion between blank and zero values, evidence from the wrong period, unresolved conflicts and claims without adequate support. This dissertation evaluates an implemented evidence-first workflow that addresses these risks through explicit data rules, traceable source records and a separately recorded verification decision. It asks how the workflow preserves identity, reporting time, missing states, sources and approval boundaries (RQ1), and what changes when a separate verification stage controls which claims enter the report (RQ2).

The prototype is a local reporting application that links each claim to its source, period, missing-data reason and approval. It separates planning, identity checks, collection, extraction, normalisation, verification, composition and approval into recorded roles with limited authority. Recorded hand-offs show who or what produced, checked and approved each statement. The roles do not imply that several agents are generally better than a simpler system. A controlled public-web route was also tested with fake providers and local fixtures, but not against a live company or website.

The evaluation used fourteen labelled fictional cases and repeated the comparison three times. Without the separate verifier, the baseline produced eleven claims: five were supported and six were not. With verification, the workflow produced the five supported claims and held all six unsupported candidates. In metric terms, precision (the share of produced claims that were supported) increased from 0.455 to 1.000, while recall (the share of required supported claims retained) remained 1.000. The combined F1 measure increased from 0.625 to 1.000, source-record completeness increased from 0.818 to 1.000, and the unsupported-claim rate fell from 0.545 to 0.000. The verifier retained an observed zero with matching evidence and held contradictory, stale and untrusted cases. These results establish the programmed mechanism on designed cases.

The contribution is therefore an inspectable evidence-first implementation together with the observed effect of its verification gate on designed cases. Because the cases and the rules were written together, the result shows that a programmed gate behaves as specified and not that the system performs well on real companies, saves analyst time or is ready for production. Manual reporting, participant review, live-web evaluation and a managed-platform comparison all lacked the observations they required, so they are set out as a prospective pilot rather than reported as findings.

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Glossary and Acronyms

This list explains the acronyms and the specialist terms used in this dissertation. Terms marked (*project term*) have a specific meaning fixed by this project’s rules rather than a general one.

Term	Meaning
Agent	A software component given one named job in a workflow, such as collecting evidence or checking a claim.
AI	Artificial intelligence.
API	Application programming interface: a fixed way for one program to ask another for data.
C0–C3	The four planned comparison conditions: manual work, a deterministic baseline, the multi-agent workflow with verification, and that workflow plus human review.
Checksum	A short fingerprint calculated from a file’s contents. If one byte changes, the fingerprint changes, so a reader can tell whether a saved file has been altered. Also called a content hash.
CLI	Command-line interface: text commands typed into a terminal rather than clicked in a browser.
Claim	(<i>project term</i>) A structured statement about one company, one measure, one value and one reporting period, which must be checked before it may appear in a report.
Claim admission	(<i>project term</i>) The decision to let a checked claim enter the report. Refusal of claim admission withholds the claim and records the reason.
Contradicted	(<i>project term</i>) A verification outcome: eligible evidence disagrees with the proposed value, so the claim is withheld and the disagreement is kept on record.
Coverage	See <i>statement coverage</i> .
CSV, JSON, XLSX	Common file formats for tabular or structured data.
D0–D2	The three dataset tiers: D0 synthetic development cases, D1 an authorised pilot, and D2 a sealed final out-of-sample set. Only D0 produced observations.
Deterministic	Producing the same output from the same input every time, with no element of chance.
Design science	A research approach in which a working artefact is built to address a stated problem, and both the artefact and the lessons from building it are treated as the contribution.
Docker, Docker Compose	Tools that package software with its dependencies into containers so it runs the same way on different machines.
Egress	Outbound network traffic leaving a system.

Term	Meaning
Evidence item	<i>(project term)</i> A saved passage from a saved source copy, recorded with its exact location, so a later reader can find the same words again.
Fixture	A fixed, made-up test input used to check that software behaves as specified.
Held	<i>(project term)</i> The D0 fixture label for refusal or non-admission: the claim is not supported, is out of date or comes from untrusted content, so it is withheld from the report with its reason recorded. It is not a human hold.
Refusal	<i>(project term)</i> Withholding claim admission when the saved passage does not support the statement, and recording that outcome instead of rewriting the claim.
HTTPS, TLS	The encrypted protocol and the encryption layer used for secure web connections.
LLM	Large language model: a model trained to generate text, such as those behind current chat assistants.
Loopback	A network address reachable only from the same machine, so the service is not exposed to the network.
Manifest	A list that binds outputs to their sources and fingerprints, so an export can be checked afterwards.
Nulls, missing states	<i>(project term)</i> The eight narrative states are blank, explicit zero, none stated, not applicable, not reported, not found publicly, source unavailable and invalid. The implemented type system also records observed, filing-not-due, dormant and not-required.
Prompt injection	Hidden instructions placed in a web page or document that try to make a language model ignore its own rules.
Provenance	The recorded origin and handling history of a piece of data: where it came from, when it was saved and what was done to it.
RAG	Retrieval-augmented generation: retrieving documents first and then generating text from them, rather than relying on the model’s memory.
RQ	Research question.
Schema	A declared structure that data must match, used to reject malformed output automatically.
Snapshot, freeze	A dated, unchanging copy of data or results, taken so later changes cannot alter what was reported.
SQLite	A small database stored as a single local file.
Statement coverage	The share of a program’s lines that were executed at least once by its tests. It shows what was exercised, not that the behaviour is correct.
<code>store=False</code>	A setting asking an external model provider not to save the request as application state. It does not remove separate abuse-monitoring retention.
Supported	<i>(project term)</i> A verification outcome: eligible evidence matches the proposed value, period and units, so the claim may enter the report.
Synthetic data	Made-up data written for testing. It contains no real company or personal information, and results from it describe the software, not the world.
URL	Uniform resource locator: a web address.
Verification	<i>(project term)</i> A separate check, carried out after a claim has been drafted, that compares the claim against its evidence and records an outcome without rewriting the claim.

Introduction

1.1 The reporting problem

An investor holding stakes in a dozen young companies has to say, in writing and on a date, how each one is doing. The awkward part is not writing the summary but standing behind it: a figure that came from a slide deck, a funding round that was announced but never completed, or an accounts filing that had not yet been made can each turn a confident sentence into one that cannot be defended when someone asks where it came from. The difficulty is structural rather than careless, because early-stage records are themselves incomplete and delayed (Hardman & Ramírez Santos, 2023), and a statement can carry a citation and still not be supported by it (Gao et al., 2023a). That question – where did this claim come from, and does the source actually say it – is the problem this dissertation addresses.

The verified project charter describes three reporting activities: someone gathers company information, someone turns it into an executive summary, and someone decides whether that summary may be released. Table 1.1 sets out who does what. These are jobs to be done rather than assumed job titles. The legal company and the reporting date must both be clear, because being incorporated does not prove that a business is trading, and recent records may be missing simply because a filing is not yet due (Galanakis & Savagar, 2026; Hardman & Ramírez Santos, 2023).

Actor	Input	Activity and hand-off	Output	Pain point addressed
Data contributor / reporting operator	Company submission and eligible public records	Collect information and hand it to aggregation	Source records and typed observations	Mixed formats, uncertain identity, blanks and source gaps
Reporting operator	Typed observations and evidence	Aggregate and analyse, then hand a draft to review	Executive-level draft with claim links	Period mismatch, unsupported claims and hidden conflicts
Reviewer / approver	Draft, evidence, conflicts and gaps	Approve, reject or return the current version	Approved report for the downstream user	Unclear release authority and unrecorded corrections

Table 1.1. Verified portfolio-reporting process and main control point

Source note: Actor and process summary from `docs/PROJECT_CHARTER.md`. The repository evidence does not establish reporting frequency, labour cost, error rate or job title.

Portfolio reports should record funding and public support as dated events with clear sources. In the Crowdcube study, an earlier successful round may act as a signal, but its meaning still depends on the

campaign and platform (Wasti et al., 2024). UK research on public support found later differences between participating firms, but it also noted selection effects and did not claim that the support caused those differences (Mondal & Mellor, 2025; Thorne et al., 2026). A round, grant or award can support a dated fact. It cannot prove future success or provide a complete funding history.

Evidence about a company’s development comes from dated events rather than one complete profile. Crowdfunding results vary by pitch, platform, country, sector, region and time. Published samples may also include only successful pitches (Estrin et al., 2024). Research on repeat funding uses the campaign round as its unit (Wasti et al., 2024). UK funding data can connect opportunities, applications, decisions, organisations, projects and outcomes, but its coverage and links remain incomplete (Thorne et al., 2026). A report should keep each event tied to its date and source.

These limits show that reliable reporting starts with the data. A company register may be suitable for one task but not another (Krasikov et al., 2020). Even required Companies House fields can be empty, so quality rules must be linked to the intended use and checked in code (Nikiforova et al., 2020). The system must record the source, date, location and check result for each important statement, and show when evidence is missing or unclear.

The main data problem is to match different records to the correct legal company and to use only information that was available at the relevant date. Research that joined several registers used normalised names, approximate matching and manual review, but some records still could not be matched (Surak & Inkley, 2026). Newer companies may also have no accounts because the filing date has not arrived (Hardman & Ramírez Santos, 2023). The prototype waits until the company number and legal identity are clear. It keeps different types of missing data instead of changing blanks to zeros.

Text can sound convincing and still be wrong, even when it includes citations. A citation marker alone does not guarantee factual correctness, citation precision or passage support (Gao et al., 2023a). A statement may also match a source that is itself wrong (Gao et al., 2023b). Public web pages can contain hidden instructions that try to change a model’s summary, source choice or tool actions (Greshake et al., 2023). Web search is therefore used only to find possible sources. Each report claim must point to a saved source and an exact passage, and a separate check must confirm the support.

The same three properties – factual correctness, citation precision and passage support – are why a report cannot treat a fluent summary as evidence. Retrieval-augmented generation can extend search and still return irrelevant or hostile passages, and a citation marker on the resulting sentence does not repair that (Gao et al., 2023a;c). The prototype therefore keeps finding, checking, extracting and verifying as separate steps.

Human approval creates a release boundary, but it does not guarantee a better result. General guidance recommends explaining what the system can do, showing uncertainty and allowing people to correct or dismiss its output (Amershi et al., 2019). In one controlled task, some participants followed incorrect AI advice (Buçinca et al., 2021). The prototype shows conflicts, missing evidence and exact source locations, and it requires a named reviewer before export. Whether review improves outcomes remains a future pilot question (Amershi et al., 2019; Buçinca et al., 2021).

Those three problems – identity, unsupported text, and over-trusted review – are why the aim is an evidence-first workflow rather than a better writer. A system that cannot refuse claim admission can only produce more sentences. A system that can refuse them can still be wrong, but the wrongness is then a recorded decision rather than a fluent gap (Gao et al., 2023a; National Institute of Standards and Technology, 2023). Chapter 3 turns that refusal into a measurable C1/C2 difference under matched inputs, and Chapter 5 reports it on designed cases rather than on a sample of companies (Fu et al.,

2026; Ribeiro et al., 2020). That equal-input comparison is the warrant taken from the 2026 preprint, not a claim that extra roles are generally better.

1.2 Why it matters

The reporting problem matters because an untraced claim consumes monitoring attention without leaving a defensible record. Section 2.2 locates that cost in the UK equity market and in weekly portfolio work; the figures themselves are stated only there (British Business Bank, 2025; Gompers et al., 2016).

1.3 Aim, research questions and objectives

The aim of this dissertation is to design, implement and evaluate an evidence-first, role-separated workflow for early-stage portfolio reporting. The study examines how the prototype preserves evidence from intake to a reviewable report and how a separate verification stage changes claim admission in controlled cases. This follows design-science research, which links a practical problem to the development and evaluation of an artefact (Hevner et al., 2004; Peffers et al., 2007).

Primary research question

How can an evidence-first, role-separated workflow produce traceable early-stage portfolio reporting claims, and what effect does a separate verification stage have on claim admission in controlled cases? The question concerns the implemented workflow rather than one model, agent label or interface. It is answered by combining RQ1 and RQ2 with the executed engineering and D0 evidence, as required in design-science evaluation (Hevner et al., 2004; Peffers et al., 2007).

RQ1: Input-to-claim accuracy and reliability

How does the implemented workflow preserve company identity, reporting time, missing states, source provenance and approval boundaries from input to report? RQ1 is answered with implementation records, engineering tests and labelled D0 cases. The assessment covers exact identity holds, value and period semantics, source-record completeness, claim-to-evidence links and repeated outcomes (Gao et al., 2023a; Nikiforova et al., 2020).

RQ2: Effect of separate verification

What effect does a separately recorded verification stage have on supported-claim precision, unsupported claims, conflict handling, source-record completeness and repeatability in D0? RQ2 uses a paired comparison between C1, which has no separate verifier, and C2, which adds verification. Both conditions receive the same fictional cases and proposed values (Gao et al., 2023a;b).

Scope narrowing

The original contract also proposed manual C0, named-review C3, live-web and managed-platform comparisons. They require authorised participants, frozen references, live access and recorded time.

Those observations were not available, so these comparisons are not active research questions (Amershi et al., 2019; Bućinca et al., 2021). Section 6.6 is the only place those absences are discussed at length.

Research objectives and evaluation plan

Four objectives connect the questions to the study. O1 defines shared rules for legal identity, dates, missing data and source records. O2 implements those rules in a workflow that stops when required evidence is missing or invalid. O3 evaluates the implemented controls and the C1/C2 verifier difference using the fixed D0 fixture. O4 distinguishes supported, unsupported, conflicting, unavailable and missing evidence in every result (Hevner et al., 2004; Peffers et al., 2007). Why the workflow separates production from judgement is argued in Section 2.6; that argument is not repeated here.

Table 1.2 summarises this evaluation plan and the evidence currently available.

Question	How it will be answered	Main measures	Current evidence and limit
Primary question	Combine RQ1 implementation evidence with the paired C1/C2 D0 comparison.	Traceability; decision correctness; supported-claim precision; unsupported-claim rate; source-record completeness; repeatability.	Engineering checks and D0 answer the bounded question. No real-portfolio or business-benefit inference is made.
RQ1	Inspect the implemented records and test fixed identity, date, missing-data, provenance and approval rules.	Required records and states; exact claim links; source completeness; repeated outcomes.	Repository implementation, tests and D0 provide the evidence.
RQ2	Compare C1 without a separate verifier with C2 on the same cases and with the same information access.	Precision; recall; F1; unsupported-claim rate; contradiction handling; source-record completeness; repeatability.	Executed on fourteen fictional cases across three repeats.

Table 1.2. Research questions and evaluation plan

Limit: Manual, participant, live-web and managed-platform comparisons are prospective work. Investment outcomes, production readiness and complete web coverage are not tested.

Source note: Developed by the author from the fixed evaluation plan and current evidence status. It does not add a study result.

1.4 Scope and boundary

The system is a local, single-user research prototype rather than a deployed service. Restricted or internal material stays local and is never sent to public generative-AI tools (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023). The project excludes buy or sell recommendations, speculative valuations, profiles of people, automatic publication and claims of production readiness (Financial Conduct Authority, 2024; Greshake et al., 2023). Those boundaries are stated once; Section 6.6 is the only later limitations section.

1.5 Contribution and roadmap

The contribution has three parts: an evidence-first prototype, an executed controlled evaluation, and a reporting method that traces claims to sources (Hevner et al., 2004; Peffers et al., 2007). The dissertation title names the implemented role structure. It is not a claim that extra agents outperform a single program. Chapter 2 reviews the literature and argues the design alternatives, including matched-input evidence for role separation. Chapter 3 defines the executed evaluation, the evidence-scope table and the ethics boundary. Chapter 4 retells the artefact as the life of a claim. Chapter 5 reports engineering and D0 results. Chapter 6 interprets RQ1 and RQ2, specifies a prospective business pilot, and states the study's limits. Chapter 7 closes the argument. This order follows design science by linking the problem, artefact, evaluation and communication (Hevner et al., 2004; Peffers et al., 2007).

Literature Review

2.1 What this review has to establish

This review has to establish four things before the artefact is described. First, what the reporting problem is, who does it and where time, rework and error enter the current cycle. Second, why public early-stage evidence resists being consolidated into one pack. Third, what makes a reporting claim defensible, as distinct from fluent or merely cited. Fourth, who is allowed to decide: which programmed role may propose a value, which may refuse claim admission, and which named person may release a report. Design-science reviews exist to fix those criteria before construction is judged (Hevner et al., 2004; Peffers et al., 2007).

Candidate records were identified in Scopus, Web of Science, IEEE Xplore, the ACM Digital Library and the ACL Anthology. Google Scholar was used only to chase citations of already included papers. The screen was limited to English. Agent and language-model work was taken from 2022 onwards. Grey literature was admitted only for governance or market context and is labelled as such in Appendix I. A systematic open-literature pass screened 53 records and retained 46 in the synthesis matrix. This chapter also uses already-admitted UK-register, governance and vendor sources that sat outside that screen, so the dissertation's admitted citation pool is larger than the matrix N . Recording both figures is required if the search method is to be inspectable (Geburu et al., 2021; Pineau et al., 2021). The appendix records the screen. It does not make the whole bibliography a systematic review.

2.2 The reporting problem: market, users, current process

The market that needs these packs is large, and the monitoring load is already high. The British Business Bank records £10.8 billion of announced UK smaller-business equity investment across 2,048 deals in 2024, while deal numbers fell by 15.1 per cent and seed- and venture-stage activity weakened (British Business Bank, 2025). A survey of 885 institutional venture capitalists reported an average of 18 hours each week working with portfolio companies, and about 60 per cent interacted with them at least weekly (Gompers et al., 2016). Those figures describe attention, not the effect of a reporting tool. They show why a claim that cannot be traced is expensive: it consumes that attention without leaving a defensible record.

Five roles share the current cycle. A data contributor supplies company records. A reporting analyst turns those records into a draft pack. A technical operator runs collection and the local tools. A reviewer decides whether the pack may be released. A downstream user, typically an investment-team reader, uses the released pack. The names are jobs to be done rather than assumed job titles.

Guidance on human–AI work treats the reviewer as a decision authority rather than as a second writer (Amershi et al., 2019; Bućinca et al., 2021). The analyst and contributor roles exist because the underlying records are mixed, delayed and incomplete (Hardman & Ramírez Santos, 2023; Nikiforova et al., 2020).

The current process is a sequence of hand-offs, and time, rework and error enter at known points rather than as a single late failure. Collection consumes time in chasing files and public extracts; error enters as mixed formats, uncertain identity and blanks that later readers cannot interpret (Krasikov et al., 2020; Nikiforova et al., 2020). Identity resolution is the first rework loop: similar names are joined by hand, and a silent merge is the characteristic error (Surak & Inkley, 2026; Thorne et al., 2026). Period matching is the second: a figure from the wrong filing window looks complete and is wrong, especially when recent accounts are not yet due (Galanakis & Savagar, 2026; Hardman & Ramírez Santos, 2023). Drafting is where unsupported fluent sentences enter, because a citation marker can sit beside a statement the passage does not support (Gao et al., 2023a;b). Review is where unrecorded corrections and unclear release authority enter; showing uncertainty does not by itself prevent over-reliance (Amershi et al., 2019; Bućinca et al., 2021).

What the pack may say is narrower than what the market talks about. Crowdfunding results vary by pitch, platform, country, sector, region and time, and published samples may include only successful pitches (Estrin et al., 2024). Repeat-funding research uses the campaign round as its unit (Wasti et al., 2024). UK work on public support found later differences between participating firms, but it also noted selection effects and did not claim that the support caused those differences (Mondal & Mellor, 2025; Thorne et al., 2026). A round, grant or award can support a dated fact. It cannot prove future success or provide a complete funding history. Financial communications must present benefits and risks fairly, which rules out speculative valuations and any general company-quality score (Financial Conduct Authority, 2024; Mondal & Mellor, 2025).

None of those five roles is measured for hours or error rate in this dissertation. The point of naming them here is to locate where a later artefact has to intervene: identity before draft, typed missingness before a zero is assumed, a recorded refusal before a conflict is averaged, and a named approval before export. The current cycle can do each of those things by habit. It does not, on the reviewed evidence, leave a complete record of them (Gao et al., 2023a; Nikiforova et al., 2020).

2.3 Why public early-stage evidence resists consolidation

Public records can support analysis and still refuse to become one complete profile. The UK register literature disagrees productively about how much that register can carry. Galanakis and colleagues treat incorporation records as a timely and broad indicator of company creation, valuing the register for its currency (Galanakis & Savagar, 2026). Hardman reaches an almost opposite conclusion about recent data: filing rules and delays make the newest records the least complete part of the register (Hardman & Ramírez Santos, 2023). Krasikov and colleagues add a third position: open access is not the same as readiness, because a register may be fully published and still unsuitable for a given business task (Krasikov et al., 2020). The disagreement is resolvable. Timeliness holds for the fact of incorporation and fails for financial filings. A report can therefore accept register identity as authoritative while treating recency of financial records as a property to be checked per field.

Linking those records to the correct legal company is a second barrier. Funding sources without shared identifiers need careful exact or approximate matching, and errors and gaps still remain (Thorne et al., 2026). Work across several registers required name normalisation, approximate comparison and

manual checks when company or country names disagreed (Surak & Inkley, 2026). Companies House identifies incorporated companies rather than all economic activity (Galanakis & Savagar, 2026). An uncertain match should stay unresolved until a named reviewer checks it. Records with similar names must not be merged automatically.

Private-market evidence is incomplete in a different way. Kaplan and Lerner describe venture-capital data as inconsistent, with definitional differences, missing failed firms and backfilling (Kaplan & Lerner, 2016). The British Business Bank treats angel-investment evidence as partial because much activity is private and reporting is patchy (British Business Bank, 2025). Missingness must keep its meaning. Documentation should state what is missing, why it is missing and what is still unknown (Gebu et al., 2021). An empty financial cell does not automatically mean zero (Bradley et al., 2024). Completeness and validity are also separate qualities that need their own checks (Nikiforova et al., 2020). Not found publicly does not mean that something does not exist.

Time makes consolidation harder still. A report needs separate dates for when an event happened, when the information became available, when it was collected and the last date allowed in the pack. Register snapshots and filing delays show why recent facts may not have been available at an earlier reporting date (Hardman & Ramírez Santos, 2023). Incorporation and later business activity happen at different times (Galanakis & Savagar, 2026). Using information published after the reporting date creates data leakage and may make an evaluation look better than it should (Kapoor & Narayanan, 2023). A correction needs a new version and date. It must not be used to support an earlier report.

Commercial products cover related but non-equivalent workflows, and setting them beside that data-quality literature exposes the gap. Visible describes portfolio-data collection, monitoring, dashboards and investor reporting; Zapflow describes integrated deal and portfolio records; and Affinity describes relationship-centred activity capture and reports (Affinity, n.d.; Visible, n.d.; Zapflow, n.d.). Each presents consolidated portfolio figures as a solved presentation problem. Kaplan and Lerner’s account of the underlying records cannot be wished away by a dashboard (Kaplan & Lerner, 2016). No vendor page reports claim-level support accuracy or how conflicting sources are resolved. The gap is not a missing screen. It is the absence of a tested rule for when a consolidated figure may be said at all. A pipeline, a single grounded agent or a bought platform can each present a pack. None of the reviewed sources shows that presentation survives a support check when two eligible records disagree (Buneman et al., 2001; Gao et al., 2023c; Guo et al., 2024). Section 2.6 and Table 2.1 set those presentation options beside the evidence contract.

2.4 What makes a claim defensible

Data quality is a property of a dataset in use, not of the dataset alone, so accuracy, timeliness, completeness and consistency each need rules bound to a particular field and purpose (Krasikov et al., 2020; Nikiforova et al., 2020). Provenance is the matching record of where a statement came from and what was done to it. Foundational database work separates why a query result exists from where its contributing values sat (Buneman et al., 2001). That explains lineage, not whether a report is right. Documentation should record purpose, contents, processing, gaps, errors, dependencies and corrections (Gebu et al., 2021), and repeating a result needs enough data, code, steps and measures (Pineau et al., 2021). A defensible portfolio statement therefore links to its saved source, its exact place in that source, the changes made to it, the checks applied and the approval given.

2.4.1 Measure, repair or refuse: ALCE versus RARR

Checking a source and checking a claim are two different steps. The source check covers access, type, permission, date and whether a stable copy can be saved. The claim check asks whether an exact passage supports the statement. Two studies by overlapping authors take different views of what to do when that second check fails. The ALCE work treats attribution as a property to be measured, separating factual correctness from citation precision and recall so that weak support becomes visible in the score (Gao et al., 2023a). The RARR work treats it as a defect to be repaired, revising generated text after the fact so that it matches retrieved evidence (Gao et al., 2023b). Both are defensible, and the choice is consequential: repair improves the finished text but relocates the decision back inside the component that produced it. A third option, refusal, is to withhold claim admission when the passage does not support the statement, and to record that outcome instead of rewriting the text. Self-preference in model evaluators rises with self-recognition, which is a demonstrated reason not to let the same component rewrite the claim it is judging (Panickssery et al., 2024; Pineau et al., 2021). Verification is also claim-level rather than report-level. FactScore decomposes long-form text into atomic facts and scores the share supported by a knowledge source, which is the grain at which admission can be refused (Gao et al., 2023a; Min et al., 2023).

Where permission and technology allow, an accepted source should be saved as a fixed local copy with a checksum, collection time, exact passage and version. A correction should create a new version rather than replace the old one. Dataset and reproducibility guidance both support recording external resources, updates, errors and superseded versions (Geburu et al., 2021; Pineau et al., 2021), and model cards apply the same logic to versioned intended use and evaluation limits (Mitchell et al., 2019). Some sources cannot legally or technically be saved at all.

Conflicting evidence must be retained. Hallucination research distinguishes factual conflict, unverifiable claims and overstatement, linking reliable output to uncertainty and checking against trusted knowledge (Huang et al., 2023). A source link does not resolve disagreement when text conflicts with evidence or the source is wrong (Gao et al., 2023b). One citation marker cannot represent correctness, passage support and completeness (Gao et al., 2023a). AttrScore finds that support is easier to judge than contradiction. The authors summarise GPT-4 at 81–83 per cent overall accuracy on AttrEval-GenSearch. The same local pages record overall 85.1 in the zero-shot setting and 84.3 in the few-shot setting, while contradiction remains the weak class at 45.0 and 53.3 (Huang et al., 2023; Yue et al., 2023). Because contradiction detection is weak, the ledger retains conflict for a named person rather than collapsing it in software. The system must not automatically prefer the majority, newest or most fluent version.

A traceable report therefore records the whole chain: discovery, checking, capture and use, then the claim, support decision, conflict, approval and export. The design principle drawn from this section is that a verifier should check evidence without being able to rewrite the claim, precisely because correctness, support and attribution are distinct properties (Amershi et al., 2019; Gao et al., 2023a;b; National Institute of Standards and Technology, 2023). Figure H.1, in Appendix H, summarises this admission chain. Measure and repair are the two positions already tested in the citation literature. Refusal of claim admission is the untested third, and it is the mechanism RQ2 isolates (Gao et al., 2023a;b).

2.5 Retrieval as remedy and vulnerability

Retrieval-augmented generation (RAG) separates document indexing, retrieval and text generation, and the literature holds two views of it that are hard to reconcile. In the hallucination literature, retrieval is the remedy: grounding output in real documents reduces invented content that a model would otherwise produce from memory (Huang et al., 2023). In the security literature, the same mechanism is the vulnerability, because indirect prompt injection lets text hidden in retrieved pages alter a model’s summary, queries, source choices or later tool use (Greshake et al., 2023). Retrieval, in other words, fixes a problem of invention by opening a channel for a problem of influence. Surveys of the field note that retrieval may equally miss relevant material or return irrelevant passages (Gao et al., 2023c). The practical consequence adopted here is that a large language model (LLM) may help find candidate sources, while finding, checking, extracting and verifying stay separate steps that retrieved text cannot reach into.

The saved Responses API documentation describes model-managed `web_search`, inline citations, annotations, domain filters and a list of consulted sources (OpenAI, n.d.-b). These features can help plan a search and produce a list of possible sources (Gao et al., 2023c). They guarantee nothing about how many sources or citations come back, and a visible citation may still be incomplete or unsupported (Gao et al., 2023a). Returned URLs, snippets and model text stay possible sources until they pass the source checks.

Structured evidence should first be extracted with fixed, rule-based code. Unstructured public or synthetic evidence may use an optional model, but only to suggest a claim that links to an exact passage. Factual correctness, passage support and citation completeness remain different measures (Gao et al., 2023a). A correct source link also does not prove a claim when the evidence disagrees (Gao et al., 2023b). Both routes end with fixed checks of the exact passage, data structure and format, followed by separate verification. Unsupported content is rejected or left unresolved. It is never filled in from model memory.

Public content found through search must be treated as untrusted data, not as instructions. Indirect prompt injection can hide harmful instructions in searched material and change summaries, queries, source choices or later actions (Greshake et al., 2023). RAG can also receive noisy, conflicting or hostile text (Gao et al., 2023c). The prototype therefore limits tool permissions, checks each destination, separates extraction from other actions, limits redirects and requests, and stops safely when a check fails. Testing, uncertainty, separate review and safe failure are recognised risk controls (National Institute of Standards and Technology, 2023). Figure I.1, in Appendix I, separates finding a source from accepting it as evidence. A report should therefore state how much searching was permitted rather than imply that the web was searched. Context size, token limits, domain filters and source lists are all configurable, and the number of sources returned is not guaranteed (OpenAI, n.d.-b), so a repeatable method records the queries, sources, reporting date and budgets, then names what was inaccessible, missing, conflicting or unverified (Gao et al., 2023a; Pineau et al., 2021). Search may propose a source; fixed checks dispose of it; retrieved text is data, never instruction (Greshake et al., 2023; National Institute of Standards and Technology, 2023).

2.6 Who is allowed to decide

Dividing a task between programmed roles is a design choice, not an acquisition of collective intelligence. Reported systems differ in interface, role description and communication structure (Guo et al., 2024),

while AutoGen shows that the developer, not the architecture, fixes the roles, tools, control flow and stopping rules (Wu et al., 2024). This is why Table H.4, in Appendix H, assigns each role one responsibility, defined inputs and outputs and a visible failure limit, and why this section rests its case on authority rather than on role count (Hevner et al., 2004; Peffers et al., 2007). Table 2.1 summarises those architecture alternatives against the evidence contract.

Approach	What it can provide	Main trade-off for this prototype	Evidence position
Manual work-flow	Flexible human collection, reconciliation and judgement	Provenance, timing and review depend on the local procedure	No measured baseline in this study
Deterministic pipeline	Repeatable rules, typed data and inspectable lineage	Limited handling of new unstructured evidence; wrong rules repeat consistently	Used for structured processing, not evaluated as a complete alternative
Single grounded agent	One interaction path can retrieve, extract and apply a fixed support gate	Production and checking share context and permissions unless separated explicitly	Credible alternative; not compared empirically
Role-separated workflow	Narrow permissions, persisted hand-offs, failure isolation and an approval boundary	More states, messages, storage and maintenance	Selected for inspectability; D0 isolates the verification gate, not general multi-agent superiority
Managed agent platform	Configured persistent roles, tools, context and hand-offs without building every interaction surface	Provider-managed boundaries and shared computer state require separate control review; setup and maintenance effects remain unknown	Grok Bot is a future comparator; not implemented or evaluated

Table 2.1. Architecture alternatives for the reporting problem

Source note: Author synthesis from Buneman et al. (2001), Nikiforova et al. (2020), Gao et al. (2023c), Guo et al. (2024), Wu et al. (2024), Cemri et al. (2025), Dibia et al. (2024) and SpaceXAI (2026). The table compares design choices; it does not report an effectiveness comparison.

One body of work argues that named roles help. Multi-agent debate can raise factuality on reasoning benchmarks (Du et al., 2024; Guo et al., 2024). Reflexion separates an actor from an evaluator and a reflection step, provided a reliable evaluation signal exists (Shinn et al., 2023; Wu et al., 2024). MetaGPT encodes standard operating procedures as role hand-offs, and ChatDev adds explicit reviewer and tester roles in a chat-chain; both report gains on software-engineering tasks (Hong et al., 2023; Qian et al., 2024). These results support Position A: role separation can help when the task supplies a checkable signal.

Position B is the caution. Under matched inputs, at most one of six multi-agent systems in a 2026 preprint exceeded the single-agent anchor on a balanced accuracy average, while the remaining five trailed by 2.56 to 11.29 points (Cemri et al., 2025; Fu et al., 2026). The same study is the warrant for C1/C2 rather than an objection to this design: a comparison that does not hold inputs equal cannot attribute a gain to extra agents. This pipeline is not claimed to beat a single agent. Matched inputs are the only credible test, and the D0 comparison isolates a recorded refusal of claim admission rather than a count of roles (Fu et al., 2026; Peffers et al., 2007).

The clearest counterweight to any enthusiasm here is empirical. Analysis of 1,642 traces from seven multi-agent systems identified fourteen failure modes spanning system design, inter-agent alignment

and task verification (Cemri et al., 2025). That evidence deserves careful reading rather than either dismissal or alarm: much of the wider dataset was annotated with model assistance, and its coding and mathematics tasks differ from portfolio reporting. Read cautiously, it still establishes that added roles are themselves a source of failure, which makes programmed control flow, tool permissions and explicit stop rules necessary rather than optional (National Institute of Standards and Technology, 2023; Wu et al., 2024).

Human review gives a person authority to decide; it does not create new evidence. Here the applied guidance and the experimental evidence are in genuine tension. The widely cited design guidelines recommend making system limits clear, showing uncertainty and letting people dismiss or correct output, treating transparency as the route to sound human judgement (Amershi et al., 2019). Bućinca and colleagues tested something close to that prescription and found it insufficient: participants still followed incorrect advice, and the interventions that reduced over-reliance did so by making the task deliberately harder, with uneven effects across participant groups (Bućinca et al., 2021). The implication is that the remedy the guidelines propose is the one the experiment found wanting, so a well-designed review screen cannot be assumed to produce better decisions. This project accordingly treats its interface as an accountability mechanism rather than an accuracy mechanism. It shows the claim, exact support, conflicts, uncertainty and content hash so that a named reviewer’s decision is recorded and attributable, and it leaves the question of whether review improves outcomes to a study with participants (National Institute of Standards and Technology, 2023).

Managed platforms make custom orchestration itself look optional, which is the sharpest challenge to this project’s design. AutoGen Studio already exposes models, tools, memory and workflows through a visual interface, so much of the wiring described above can be bought rather than written; its authors nonetheless call the tool not production-ready and name architecture comparison and reliability as open problems (Dibia et al., 2024). Vendor documentation for Grok Bot goes further, describing persistent named Bots with distinct jobs, tools, working rules and approval boundaries (SpaceXAI, 2026). The unresolved question is not whether a platform can host roles, but whether it can enforce an evidence contract. Convenience in configuring agents is a different property from correctness in admitting claims, and neither source measures the second (Dibia et al., 2024; Guo et al., 2024). Evaluating such a workflow therefore means keeping its failures visible rather than reporting only its successes. Logs should carry role, task, attempt, input, output, hash, time, tokens, errors and recovery, and fixed cases should cover each role and the whole workflow, including disagreement and safe failure (Guo et al., 2024; National Institute of Standards and Technology, 2023; Pineau et al., 2021). The benefit of role separation is therefore contested; matched inputs are the test; this pipeline is not claimed to beat a single agent (Cemri et al., 2025; Fu et al., 2026).

2.7 Settled / contested / gap

Three findings are settled enough to bind the design. A reporting claim needs a legal identity, a reporting date and a saved source; citation is not the same as support; and conflict should be retained rather than averaged (Buneman et al., 2001; Gao et al., 2023a; Kaplan & Lerner, 2016). Two findings remain contested: whether extra programmed roles improve outcomes under matched inputs, and whether human review improves decisions rather than merely recording them (Amershi et al., 2019; Bućinca et al., 2021; Fu et al., 2026). The gap this dissertation addresses is narrower than a missing product. No located study tests whether a separately recorded verification stage changes what a report may say when inputs are held equal, or whether evidence survives from intake to approved export. Fu

and colleagues’ matched-input result is why a manual baseline is postponed to the pilot in Section 6.5 rather than treated as an omitted control (Fu et al., 2026; Peffers et al., 2007).

For RQ1, the established problem is that company and funding records can be incomplete, delayed and hard to link, while an exact citation can still sit beside a wrong or partial statement (Gao et al., 2023b; Kaplan & Lerner, 2016; Thorne et al., 2026). What remains unresolved is what survives the whole chain. RQ1 therefore needs labelled outcomes at each step, not a demonstration that a report can be produced (Nikiforova et al., 2020; Pineau et al., 2021). For RQ2, splitting the work between named roles can make responsibilities visible, but surveys and trace analysis also find failures of coordination, verification and system design; one program applying the same fixed gate remains a credible alternative (Cemri et al., 2025; Guo et al., 2024). What remains unresolved is whether a separately recorded verifier changes which claims are admitted when the inputs are equal. Table 2.2 maps those positions to the questions Chapter 3 will operationalise. How that instrument is built, and why a designed suite is used instead of a field rate, is a methods question rather than a literature finding (Hevner et al., 2004; Peffers et al., 2007). The supporting review-source, signal-boundary and data-semantics tables are provided as Tables H.1, H.2 and H.3 in Appendix H.

Status	Position	Where it is taken
Settled	A claim needs legal identity, reporting date and a saved source.	RQ1; Chapter 4 stages 1–3
Settled	A citation marker is not the same as passage support.	RQ2; Chapter 3 measures
Settled	Conflict should be retained, not averaged or newest-wins.	Chapter 4 stage 6; Chapter 5 cases
Contested	Extra programmed roles improve outcomes.	Section 2.6; D0 isolates the gate, not role count
Contested	Human review improves decisions rather than only recording them.	Section 6.5
Gap	Matched-input effect of a recorded refusal of claim admission.	RQ2; D0 C1 versus C2
Gap	Live-source, manual and participant comparisons.	Table 3.1, planned only

Table 2.2. Settled findings, contested claims and the research gap

Source note: Author synthesis of the Chapter 2 argument. The table is a reading guide, not a result.

Research Design and Methodology

3.1 Design-science strategy and what counts as evidence

This dissertation uses an applied design-science strategy because the research object is a working portfolio-reporting artefact for a bounded organisational problem. Design science links construction to evaluation against stated objectives (Hevner et al., 2004; Peffers et al., 2007). The method therefore separates three kinds of evidence: implementation records show what was built, engineering tests show which specified behaviours were exercised, and D0 shows how two workflow conditions handled labelled fictional cases.

The student built and evaluated the artefact, which creates confirmation risk. The main controls were fixed case labels and metrics, versioned code and fixtures, saved result files, complete failure accounting and repeated runs (National Institute of Standards and Technology, 2023; Pineau et al., 2021). These controls make changes visible. They do not provide independent validation.

A small set of designed cases is easily mistaken for a weak substitute for a large sample. Ribeiro and colleagues argue the opposite for behavioural testing: held-out accuracy on a single test set overstates real capability, and a structured suite of targeted cases, each built to probe one specified behaviour, exposes failures that aggregate scores conceal (Ribeiro et al., 2020). That is the warrant for D0. Kapoor and Narayanan show how easily designed or leaky setups flatter a system (Kapoor & Narayanan, 2023). Read together, the two support a diagnostic rather than estimative reading: passing D0 shows that a targeted control is present.

Figure 3.1 shows the evidence boundary used throughout the study. The figure's reporting stage is labelled for RQ1–RQ2, which are the only active questions. Manual work, participant review, live-web operation and managed-platform comparison were removed from the active research questions because they had no authorised observations (Hevner et al., 2004; National Institute of Standards and Technology, 2023). Table 3.1 is the single downstream reference for that cut.

3.2 Design decisions and rejected alternatives

I chose fourteen cases across thirteen designed categories, with two probes in the blank-versus-zero category, because the question is whether a specified gate fires, not what rate it would achieve on a population of companies. A further case in the same category would still be a designed probe. It would not become a sample. Ribeiro's checklist logic is the reason for that shape (Kapoor & Narayanan, 2023; Ribeiro et al., 2020). The categories are complete and sparse records, blank against observed

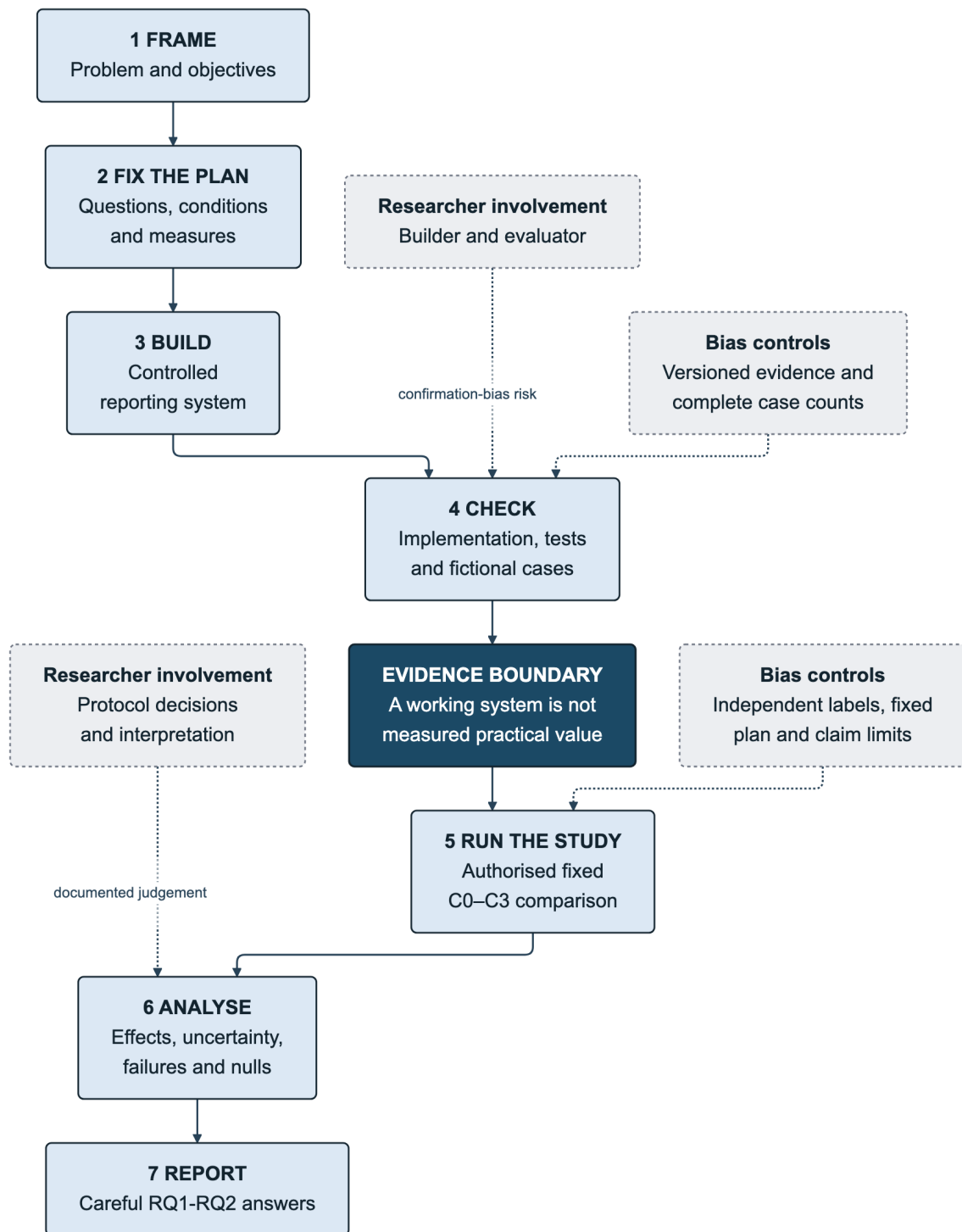


Figure 3.1. Boundary between construction, engineering checks and study results. Developed by the author from the cited methods literature and the project's fixed rules.

zero, mixed types, conflicting and stale evidence, unsupported and inaccessible sources, unclear identity, duplicate support, narrative evidence, mixed currencies and prompt injection. A single designed probe per remaining category is enough to ask whether the named gate fires. The blank-versus-zero pair is the exception, because collapsing those two states is the error the catalogue exists to prevent. That shape is still not enough to say how often a gate would fire on a portfolio, which is why Chapter 5 reports case decisions beside the rates (Pineau et al., 2021; Ribeiro et al., 2020).

I chose synthetic material because it supplies controlled labels. Bradley and colleagues show that empty cells, zeros and table positions are easy to confuse in financial extraction, which is exactly the distinction D0 needs to hold visible (Bradley et al., 2024; Gebru et al., 2021). A real company panel would have been preferable for transfer. It was not used here because independent labels, a sealed freeze and authority to process restricted portfolio data were not available. That absence is sequenced in Section 3.6, not treated as a silent omission.

Three repeats are there to test exact repeatability of a programmed gate, not to estimate variance in a sample. Reproducibility guidance asks for enough procedure that another person can inspect the route to a result (Mitchell et al., 2019; Pineau et al., 2021). The first repeat is scored. The complete ordered outcome list is compared across all three. Agreement is therefore a property of the fixture and the code, not a confidence interval.

Precision, recall and F1 are used because RQ2 is about claim admission, not about cost or speed. Precision asks what share of emitted claims were supported. Recall asks whether required supported claims were kept. The unsupported-claim rate uses the same emitted-claim denominator as precision. Cost-weighted scores would have mixed a mechanism question with an efficiency question that D0 cannot answer, because both routes are local fixed rules with truncated sub-millisecond times (Gao et al., 2023a;b).

C1 versus C2 is the comparison because Fu and colleagues show that extra agents are not a result until inputs are held equal. Under matched conditions, at most one of six multi-agent systems beat the single-agent anchor (Fu et al., 2026; Guo et al., 2024). That preprint is the warrant for giving C1 and C2 the same cases, catalogue, candidate values and evidence, and for refusing to read the difference as a verdict on multi-agent design. What differs is a separately recorded verification stage that can refuse claim admission. Composition cannot override that decision (Peffer et al., 2007; Wu et al., 2024).

I also rejected a third D0 cell that would have pretended to be a manual baseline. Writing a human-looking score beside the same fixture would have looked like C0 without being one. Fu and colleagues' matched-input requirement applies to that temptation as well: a fake manual cell would not have held effort, sources or stopping rules equal (Fu et al., 2026; Kapoor & Narayanan, 2023). The honest C0 belongs in the authorised pilot, where a person and the workflow see the same company-periods.

The evaluand is the complete local workflow configuration: input rules, catalogue, source access, operational roles, verification, prompts, schemas, computing environment and stop conditions. Design-science evaluation requires comparison with stated objectives, and system reporting should state the intended use, version and evaluation procedure (Hevner et al., 2004; Mitchell et al., 2019). The original protocol also defined C0 manual work and C3 named review. Those conditions did not run. Their detailed condition matrix is kept as Table D.1 in Appendix D, so the planned comparison remains inspectable without presenting missing observations as evidence (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

For D0, both conditions receive the same candidate value, evidence record, reporting period and precondition. Scoring uses the fixture's expected emission and status fields. The evaluator does not give

either condition later evidence or a wider source window. This equal-input design is sufficient for the narrow C1/C2 mechanism comparison. The visible development labels are not an independent reference (Kapoor & Narayanan, 2023; Pineau et al., 2021). A real-company evaluation would require an eligible-source list, cut-off date, independently labelled claims and recorded adjudication, and agreement would indicate consistency rather than truth because reviewers can share an error (Artstein & Poesio, 2008; Gao et al., 2023a).

3.3 What was validated and what was not

Table 3.1 classifies every downstream claim in this dissertation as validated, built but not evaluated, or planned only. Engineering tests and D0 sit in the first column. The public-web route, live official connectors and live company-research sit in the second. Manual work, named review, sealed evaluation and the business pilot sit in the third (Hevner et al., 2004; Pineau et al., 2021). Later chapters cite this table rather than restating the cut.

Validated in this study	Built, not evaluated	Planned only
Local identity, missing-state, period, provenance, verification, approval and export controls. Paired C1/C2 D0 on fourteen labelled fictional cases, three repeats. 286 automated tests on the 28 August 2026 snapshot. Offline connector replay and controlled adversarial tests with fake providers.	Live public-web discovery, capture and extraction. Direct live Companies House and UKRI retrieval (held at gate G2). External-model extraction on real companies (only synthetic smoke runs exist). Live company-research against an open website.	Manual baseline C0. Named-review condition C3. D1 independent labels and sealed D2. Managed-platform comparison. Accessibility and independent security assessment. Authorised business pilot.

Table 3.1. Evidence scope for this dissertation

Source note: Author classification of the executed snapshot, the repository and the unrun protocols. Later chapters cite this table rather than restating the boundary.

The three columns are mutually exclusive for any given claim. A passing engineering test cannot be read as a live-source result, and an unrun protocol cannot be read as a negative finding (Hevner et al., 2004; Pineau et al., 2021). The left-hand column is the only column Chapter 5 may treat as a result. The middle column is why Chapter 4 can describe a public-web funnel without Chapter 5 reporting a live-web accuracy: the code exists, the specified control fired on fake providers, and no authorised live endpoint was contacted. The right-hand column is why interviews, a manual baseline and a usability study are absent from the results rather than reported as zeros (Fu et al., 2026; Kapoor & Narayanan, 2023). Reading down the table is therefore a reading rule for the rest of the report. Later chapters point back to it instead of restating that cut. The planned-only column is also why this chapter does not invent a substitute observation. An interview summary, an expert rating or a usability score that was never collected cannot be inferred from the existence of a review screen (Amershi et al., 2019; Buginca et al., 2021). The table is the place those absences are named once.

3.4 D0 fixture

D0 contains fourteen visible, labelled fictional cases. The fixture and its evaluation record are checksum-pinned. Loading checks the schema version, data classification, source version, file checksum and test-only namespace. Case and evidence identifiers sit in a benchmark namespace that the operational collector rejects. The complete case ledger is Table F.2 in Appendix F (Gebru et al., 2021; Pineau et al., 2021).

Error-detection and error-injection studies give the same warrant as Ribeiro. Detectors on ReaLMistake run at very low recall compared with humans, whose agreement reached an F1 of 95.7 (Kamoi et al., 2024; Ribeiro et al., 2020). On SummExecEdit the best joint detect-and-explain score was 0.49, and 60 inconsistent summaries were missed by every model tested (Kamoi et al., 2024; Thorat et al., 2024). D0 is therefore written as labelled refusals of claim admission, including the fixture outcome held, which this dissertation uses as a refusal or non-admission label rather than as a human hold.

D0 is a development fixture. The rules and cases were written together, so the result establishes only whether the programmed gate behaves as designed on these cases. A future pilot must create independent labels before a sealed final set is opened (Kapoor & Narayanan, 2023; Pineau et al., 2021).

Error analysis uses the expected verification states and records false emission, false withholding, conflict, stale evidence, inaccessible evidence, identity hold and untrusted content separately. Negative and unavailable outcomes remain in the record. Any later study should pair the same company-periods, report exclusions before rates and use uncertainty at the company or report level, not treat repeated fields as independent observations (Kapoor & Narayanan, 2023; Pineau et al., 2021).

3.5 Measures and denominators

The primary D0 measures are supported-claim precision, recall, F1, unsupported-claim rate, source-record completeness and exact repeat consistency. Table 3.2 states the denominators used in Chapter 5. Contradiction accuracy has two applicable cases and is read as a small case result, not a stable rate. Duration values below one millisecond were truncated to zero. Model cost is zero because D0 uses fixed local rules. Measures without D0 labels remain null, not zero (Gao et al., 2023a; Nikiforova et al., 2020).

Measure	Numerator / denominator	Why it is used
Supported-claim precision	Supported emitted claims / all emitted claims	Does refusal raise the share of admitted claims that are supported?
Recall	Correctly emitted required claims / all required claims	Does refusal drop claims that should have been said?
Unsupported-claim rate	Unsupported emitted claims / all emitted claims	The complement of precision on the same emitted set.
Source-record completeness	Complete source records / emitted claims	Completeness of the record, not discovery of new sources.
Exact repeat consistency	Identical ordered outcomes across three repeats	Repeatability of the programmed gate.

Table 3.2. Primary D0 denominators

Source note: Compact view of the metric matrix in Appendix F. Cost, time and reviewer measures have no D0 labels and remain null.

The complete metric matrix in Appendix F also defines prospective human time, edit, usability and cost measures. Those fields are pilot metrics, not results of this study (National Institute of Standards and Technology, 2023; Pineau et al., 2021). No inferential test is reported, because D0 is not a sample (Kapoor & Narayanan, 2023; Ribeiro et al., 2020). The evaluator reports counts before rates and keeps all fourteen paired cases in both conditions. Individual cases explain why an aggregate changed. Schema and normalisation accuracy are also reported because both conditions share earlier processing, so a difference there would not be a verifier effect (Gao et al., 2023a; Pineau et al., 2021). Contradiction accuracy uses two applicable cases. Reporting it as 0.000 versus 1.000 is honest about those two cases and misleading if read as a stable detector rate. Duration and model cost stay in the record as truncated zeros so that a later reader can see they were collected and were uninformative, rather than quietly dropped (Gebru et al., 2021; Nikiforova et al., 2020).

3.6 Scope decisions: no interviews, no manual baseline

Interviews, expert ratings and a stakeholder usability study were not run. They would have required authorised participants, a frozen protocol and a recorded coding frame. Those observations do not exist, so they are not inferred from the interface (Amershi et al., 2019; Buçinca et al., 2021). Promised-but-unrun interview or rating work is therefore absent from Chapter 5. It is not reported as a weak result or as a zero score.

A manual baseline was postponed for the same sequencing reason, not as an apology. Fu and colleagues show that a comparison that does not hold inputs equal cannot attribute a gain to extra machinery (Fu et al., 2026; Peffers et al., 2007). A C0 cell that used different sources, a different cut-off or a different completion rule would not isolate the verifier. I did not run interviews because they would have answered a different question: how operators talk about reporting, not whether a programmed refusal changes claim admission under matched inputs. A usability study of the review screen would likewise measure preference and effort, not support (Amershi et al., 2019; Buçinca et al., 2021). The matched manual comparison therefore belongs in the authorised pilot in Section 6.5, where the same company-periods, source window and stopping rule can be frozen before anyone sees the outcomes (Kapoor & Narayanan, 2023; Pineau et al., 2021).

3.7 Ethics, data classification, reproducibility

Every input is classified as restricted, internal, public or synthetic. Restricted material remains local. Public availability does not itself authorise every use (Central Digital and Data Office, 2023; Gebru et al., 2021; National Institute of Standards and Technology, 2023). D0 uses only fictional data and no external source or model call. Local copies are checksum-linked, written once where required, kept out of Git and privately permissioned. Personal contact details are removed before an accepted public page is stored.

The optional external route sends only approved public or synthetic company-level material. It requires open collection and model gates, reviewed identity, a named reviewer, an approved route and a runtime key. Requests use `store=False`, strict schemas and fixed limits. Bounded synthetic smoke runs were executed on that optional route. The run manifests are not in this tree. The current code pins `gpt-5.6-luna`. Those runs are connectivity evidence, not a result. No restricted, internal or real-company material was sent. OpenAI separates saved application state from abuse-monitoring retention, and `store=False` is not Zero Data Retention (Central Digital and Data Office, 2023; OpenAI, n.d.-a).

The evaluation record retains the Git revision and dirty-state warning, platform and dependency versions, migration version, fixture and protocol checksums, test commands, result files, failures and output hashes. Collection stops when authority is unclear, a credential is exposed, unexpected personal data appears, source terms would be breached, restricted data would reach an unapproved service, or a sealed evaluation set is compromised (National Institute of Standards and Technology, 2023; Pineau et al., 2021). The report links company events to evidence. It does not advise buying or selling, rank companies, or profile individuals (Central Digital and Data Office, 2023; Financial Conduct Authority, 2024). These controls define the research boundary. They do not imply ethics approval for the future pilot.

Public pages may contain hidden instructions, misleading identities or malformed content. The route treats downloaded text as data, limits model tools, rejects instruction-like passages and demands exact support before it accepts a claim. The tests cover chosen risks; the controls cannot spot every hostile form (Greshake et al., 2023; National Institute of Standards and Technology, 2024). Collection checks the protocol, embedded credentials, ports, network addresses, redirects, robots rules, content type, elapsed time and decompressed size, and connects to a verified public address while still checking the certificate hostname. Controlled tests covering internal-service access, address switching, redirects, content type and size all passed. Each test confirms that one specified input was refused (National Institute of Standards and Technology, 2023; 2024). Budgets, bounded retries, cancellation, task fingerprints, checksums and staged export together limit runaway cost, stale completion, altered files and partial writes. A forced stop can still leave work that needs deliberate recovery (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

Work with real restricted data or with participants stays held until an approved data-management and ethics plan names who controls the data and who owns purpose, access, retention, deletion and incident response. No retention period has been invented in the meantime. What the prototype demonstrates is a local purpose limit, which is a design property rather than a data-protection judgement (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023). The research concerns companies, not individuals. Captured contact details are removed; director, employee, investor or journalist profiles, inferred sensitive traits and reputation scores are excluded. A named public statement may support a relevant company claim but cannot become a judgement about that person. Human approval cannot expand these limits (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

The Artefact: the life of a claim

4.1 Requirements and trust boundary

This chapter follows one claim from intake to approved export. The release is a local, single-user research prototype. It listens only on the machine that runs it and refuses any client that is not local. A bounded company-research route may start from a Companies House number alone, but a named person must review that number before research can begin. Incorporation is not proof that the company trades (Galanakis & Savagar, 2026; Hardman & Ramírez Santos, 2023).

Every item is labelled public, internal, restricted or synthetic. External-model processing is off by default and, when switched on, accepts only public or synthetic evidence. The client sends `store=False` with every request, which controls saved application state but not abuse-monitoring retention (Gebru et al., 2021; OpenAI, n.d.-a). Export is blocked until the current version is approved. Table 4.1 states the trust boundary. The full requirement matrix is Table C.1 in Appendix C (Financial Conduct Authority, 2024; National Institute of Standards and Technology, 2023).

Boundary	Control in the prototype	Status
Local runtime	Loopback dashboard, private API, SQLite, ignored local files; non-local clients refused.	In place; not production sign-in.
Identity-first intake	A well-formed Companies House number is an identity claim until a named reviewer accepts it. Names do not merge records.	In place; live register held.
Data class	Restricted and internal material stays local. Public or synthetic material may use the external route only when gates are open.	Control implemented; legal authority not verified.
Review and export	Named reviewer, reason and version; edits cancel approval; no export before current approval.	In place offline; human outcome unavailable.

Table 4.1. Trust boundary of the local prototype

Source note: Compact view of Table C.1 in Appendix C. The full matrix maps each control to residual risk.

4.2 Architecture in one page

The artefact runs as two local processes. Next.js renders the operator dashboard. Its server-side proxy calls a private FastAPI service, so the browser never reaches the API directly. SQLite holds identity, workflow, claim, review and audit records. Imported bytes and source captures are written once into local directories that Git ignores, verified by checksum and recorded by path (Geburu et al., 2021; Pineau et al., 2021). Docker Compose mirrors that two-process shape in pinned images. Figure 4.1 maps the request and data path (Hevner et al., 2004; National Institute of Standards and Technology, 2023). The canonical data model, including the separate company-research track, is Figure C.1 in Appendix C. Startup migrates the database to the declared head rather than creating tables directly, so the schema always has a stated version. Alembic revisions 0001–0009 reproduce the evaluated schema; revision 0010 was added after the dated snapshot, which Chapter 5 records at 0009. Tests compare the migrated schema against the one the code expects (Geburu et al., 2021; Pineau et al., 2021).

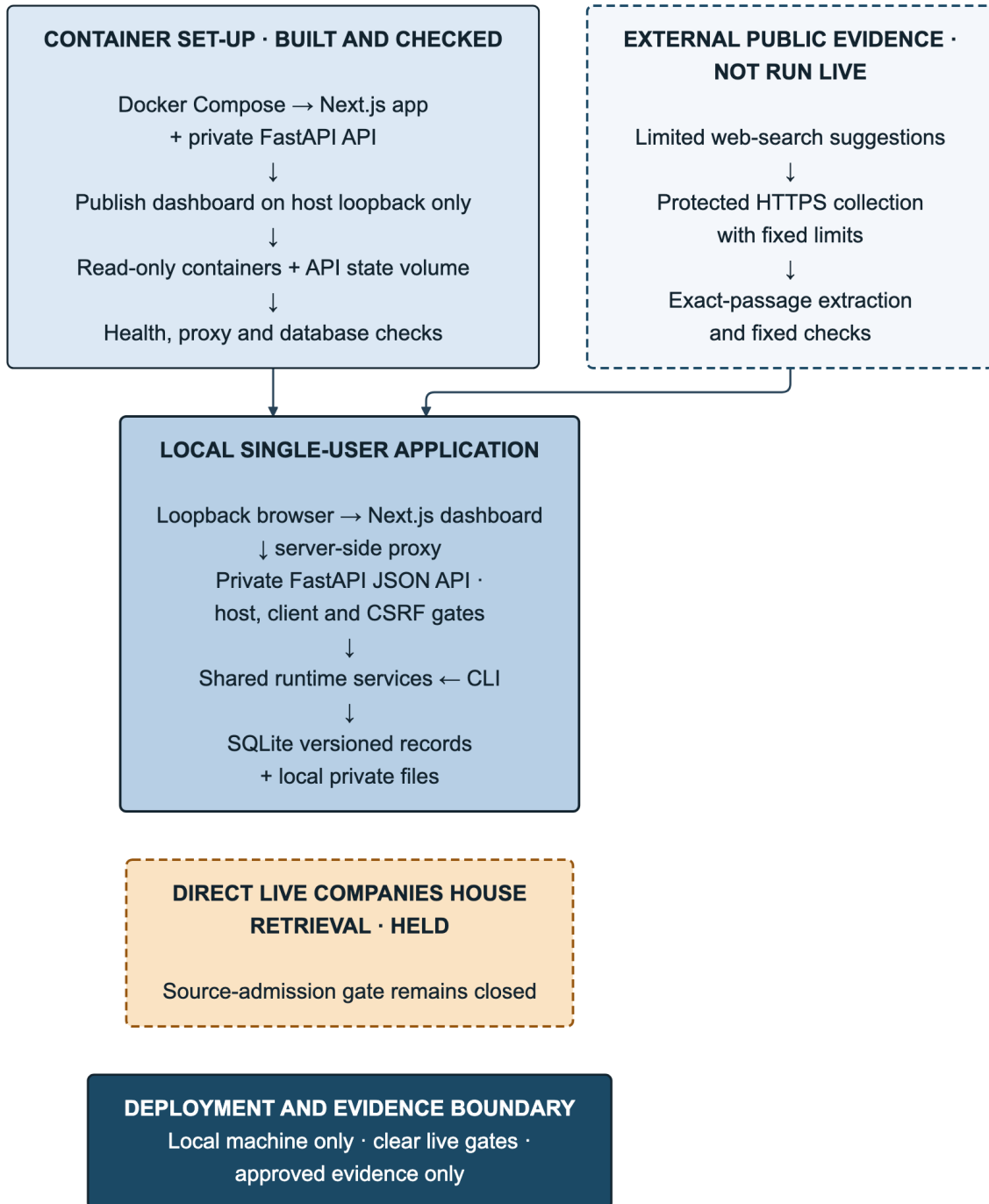


Figure 4.1. Local architecture, deployment and external-boundary status. Author synthesis of the repository’s contracts, its implementation and the local verification evidence.

4.3 Stages 1–3: intake, identity and capture

The workflow is a fixed state machine with eight stages: plan, resolve identity, collect evidence, extract values, normalise them, verify claims, compose a report and stop at human review. Each stage leaves a persisted run record holding input and output hashes, status, timing and attempts. A failure stops the stages that would have followed it (Hevner et al., 2004; Pineau et al., 2021).

Identity is stage 2. The portfolio-import route accepts XLSX, CSV and JSON, hashes the original bytes and writes a create-once snapshot. Every company row passes through the same identity resolver. A conflicting identifier becomes a counted hold rather than a silent merge. A well-formed Companies House number creates an exact identifier and a held case. It is not confirmed against the register, because valid shape is not proof of existence. Names may suggest a candidate. They never merge companies (Surak & Inkley, 2026; Thorne et al., 2026). Identity becomes ready only after a named actor accepts the exact identifier with a substantive reason. Figure C.2, in Appendix C, maps these gates.

Capture is stage 3. Collection stores run-linked evidence and its time decision before extraction. Only temporally eligible, trusted public evidence reaches later stages. Untrusted material is counted and bypassed. Connector snapshots record source and version, reviewed identifier, cutoff, locator, media type, checksum, retrieval and publication times, and classification (Gebu et al., 2021; Pineau et al., 2021). Direct live Companies House and UKRI retrieval remain held at gate G2. Offline replay of those adapters is implementation evidence, recorded in Appendix C.

Time is judged relative to the claim, not to the moment of collection. Evidence is ineligible if it carries no recorded publication or availability time, was published after the reporting cut-off, takes effect in the future, or had expired before the period began. When it was retrieved is tracked separately. A cumulative figure must span exactly from the stored programme start to the cut-off. UKRI funding binds only when every award in the window carries a finite, explicit sterling amount, and otherwise leaves a missing state with a diagnostic rather than an inferred zero (Hardman & Ramírez Santos, 2023; Kapoor & Narayanan, 2023; Nikiforova et al., 2020). The source layer is a registry of declared capabilities rather than a crawler. Every request names the source, company, reviewed identifier, cut-off, fact keys and mode, and the registry refuses it before any adapter runs if the capability is unsupported or the identity is unresolved. A refusal of this kind is a decision not to look, recorded differently from a publisher having no record (Gebu et al., 2021; Pineau et al., 2021). Two deterministic adapters are usable through synthetic offline replay: Companies House (manifest 1.4.0) matches an exact number, and UKRI (manifest 1.3.0) matches an exact organisation identifier and selects the most recent correction available at the cut-off (Galanakis & Savagar, 2026; Hardman & Ramírez Santos, 2023).

4.4 Stages 4–5: extraction and normalisation

Stage 4 extracts a candidate value. Structured evidence is extracted with fixed, rule-based code. A strict schema binds company, metric, period, value and exact evidence locator. Unstructured public or synthetic evidence may use an optional model only to suggest a claim that still links to an exact passage (Gao et al., 2023a;b). Existing extractions are reused. Every provider attempt is stored, including the failed ones.

Stage 5 normalises type, unit, currency and missing state against a versioned catalogue. Blank, explicit

zero, none stated, not applicable, not reported, not found publicly, source unavailable and invalid are the eight states used in the narrative. Distinguishing them is what allows a later reader to see why a figure is absent, and what allows stage 6 to treat an observed zero as sayable rather than as a blank to be dropped (Bradley et al., 2024; Gebru et al., 2021; Nikiforova et al., 2020). The implemented type system also records observed, filing-not-due, dormant and not-required. Many of these semantic states are stored as strings or JSON and enforced by application code rather than by the database, so the schema alone cannot guarantee they are used consistently. The catalogue version is part of the evaluand: a later change of missing-state rules is a new configuration, not a silent improvement of the same one (Mitchell et al., 2019; Pineau et al., 2021).

4.5 Stage 6: separate verification

Verification is the stage that decides what may be said. Before a claim is constructed, the workflow checks sourceability, provenance, period, trust, value and currency, then writes a verification record beside the claim. The verifier is a pure function returning one of five states: supported, contradicted, insufficient evidence, stale, or rejected as untrusted. A conflict between two current public sources cannot be averaged away. The verifier runs separately from the extractor and the composer and cannot rewrite their records, so a claim can be refused but not quietly repaired (Gao et al., 2023a;b; National Institute of Standards and Technology, 2023). This is separation of duties inside one program, not independent judgement by a second party.

Figure 4.2 is the anchor for that contract. A single program could apply the identical support rule. The D0 comparison isolates the rule and its authority over composition, not the number of roles (Guo et al., 2024; Peffers et al., 2007; Wu et al., 2024). Claims persist normalised values and link many-to-many with evidence items; each evidence item carries publisher, locator, times, checksum, connector version, classification and trust or time flags. A separate verification stage records supported, contradicted, insufficient, stale or untrusted outcomes without rewriting the source (Gao et al., 2023a;b). Portfolio reports can export these links without raw evidence. Where two sources disagree, the company-research track groups the statements as contradiction candidates for a named person to resolve. No live company has been run through that path (National Institute of Standards and Technology, 2023; Yue et al., 2023).

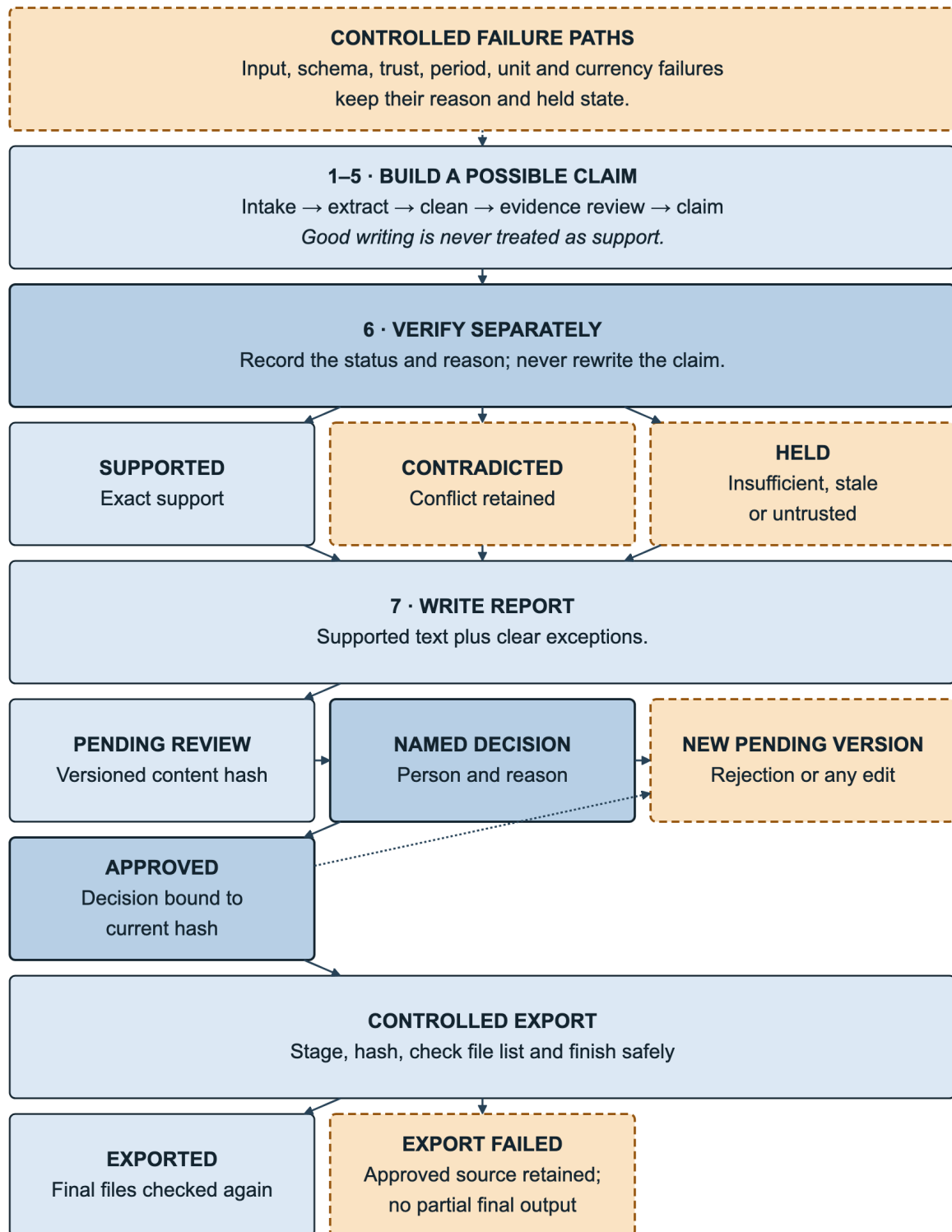


Figure 4.2. Fixed workflow, separate verification states, named approval and controlled export. Author synthesis of the implemented local controls.

4.6 Stages 7–8: composition, approval and export

Composition places supported claims in the narrative and exposes other states as exceptions. It hashes the current, versioned report and moves it to pending review. A named reviewer may approve, reject or request an edit. Each decision records actor, reason and report version. Editing creates a new section version, recalculates the content hash and revokes approval. Export remains unavailable until current approval (Amershi et al., 2019; Bućinca et al., 2021; Pineau et al., 2021). Files are staged below the local export root, bound to a versioned manifest and content hash, then re-verified before reuse. The system writes files to a folder. It does not publish.

Company-research profiles use a separate, narrower gate. Before a named approve or reject decision, the service rechecks the profile hash, pinned run contract and completed composition task; stale locks, altered content and unfinished recovery are refused. Contradiction candidates and coverage gaps stay inside the dossier the reviewer sees. Only approved profiles can be downloaded, as JSON or printable HTML (Gao et al., 2023a; National Institute of Standards and Technology, 2023). Reports are versioned by appending rather than overwriting, so an edit retires the current section instead of erasing it. Approval is bound to an exact version of exact content: change the content and the approval falls away (Amershi et al., 2019; Pineau et al., 2021).

4.7 Two claims traced end to end

Two D0 cases show what these stages do to a single claim. Both use fictional companies, a pinned fixture and the same candidate-preparation route. C1 emits eligible cleaned values. C2 admits a claim only when the separate verifier marks it supported. Chapter 5 reports the aggregate rates. This section shows the decisions themselves (Bradley et al., 2024; Gao et al., 2023a).

The first claim is an observed zero: jobs created equals 0. The submitted figure is zero. The stored evidence also says zero. The two agree, so the verifier marks the claim supported and it appears in the report with its source attached. The obvious shortcut would be to treat a zero as an empty box and quietly drop it. Doing so would remove a real and possibly unwelcome fact. The eight narrative missing states exist so that an observed nothing remains sayable (Bradley et al., 2024; Gebru et al., 2021).

The second claim is a contradiction: grant funding of £100 against register evidence of £200 for the same period. The two figures cannot both be right, and nothing in the evidence establishes which is. The verifier marks the claim contradicted. The report omits it. The disagreement is kept on record for a named person. The system does not choose the larger figure, the newer one or an average, because any of those would manufacture a number that no source states (Gao et al., 2023a;b; Yue et al., 2023). For a reader, the useful contrast is that C1 emitted this claim while C2 refused admission of it. Table 5.3 in Chapter 5 records both outcomes against the fixture locators.

Table 4.2 walks both rows through the eight stages: intake, identity, capture, extract, normalise, verify, compose, and review or export. The locators are `submission://synthetic#B2` for the observed zero and `fixture://conflict` for the grant pair. The weekly monitoring load in Section 2.2 is why those locators have to survive: an untraced figure spends attention without leaving a defensible record (British Business Bank, 2025; Gompers et al., 2016). The walk is therefore a reading guide for those two fixture rows, not a second scored evaluation of the same fixture.

At intake the operator records the candidate as typed. The observed-zero row arrives as jobs created

Stage	Observed zero (<code>zero_is_observed;</code> <code>submission://synthetic#B2</code>)	Public-value conflict (<code>conflicting_public_value;</code> <code>fixture://conflict</code>)
Intake	Candidate arrives as jobs created = 0.	Candidate arrives as grant funding of £100.
Identity	Fictional company; no identity hold on this row.	Fictional company; no identity hold on this row.
Capture	Portfolio submission stored at the B2 locator.	Synthetic register stores £200 for the same period.
Extract	Candidate value remains 0, bound to the locator.	Candidate value remains £100, bound to the locator.
Normalise	Typed as an explicit zero, not a blank.	Two current sterling figures kept; no average.
Verify	Submission and evidence agree; state is supported.	£100 against £200; state is contradicted.
Compose	C2 emits the claim with its source record.	C2 withholds the claim and retains the disagreement.
Review / export	An approved version may include the emitted zero.	Export cannot invent a third grant figure.

Table 4.2. Eight-stage walk of the observed-zero and grant-conflict D0 rows

Source note: Locators and values from `fixtures/evaluation_cases.json`. D0 scores the admission decision (normalise and verify). Stages 1–3 and 7–8 for these two rows are the workflow contract, evidenced by engineering tests, not a full pipeline replay of the fixture.

equals 0. The conflict row arrives as grant funding of £100. Neither figure is yet a report sentence. Identity then binds each row to one fictional legal company. These two D0 rows do not exercise an identity hold. The hold is the same contract that later keeps two similarly named companies apart, and it is evidenced by engineering tests rather than by these fixture identifiers. Capture stores the supporting bytes before anyone extracts a value. The zero is stored at `submission://synthetic#B2`. The register figure of £200 is stored at `fixture://conflict`. If those locators were missing, a later reader could not reopen the same evidence. Stages 1 to 3 for these two rows are therefore the workflow contract. They are evidenced by engineering tests, not by a full pipeline replay of the fixture through the dashboard (Geburu et al., 2021; Nikiforova et al., 2020; Pineau et al., 2021).

Extract binds company, metric, period, value and exact locator. For the first row the candidate remains 0. For the second it remains £100. Normalise is where the catalogue earns its keep. The zero is typed as an explicit zero rather than as a blank, none stated or not reported, so it stays sayable. The grant pair stays as two current sterling amounts for the same period. Nothing is converted, averaged or replaced by the newer figure. Verify then writes the admission decision. The two zeros agree, so the state is supported. The two grant figures disagree, so the state is contradicted. D0 scores that admission decision, which is the normalise-plus-verify pair. It does not score whether an operator liked the screen (Bradley et al., 2024; Gao et al., 2023a; Yue et al., 2023).

Compose is allowed to write only what verify admitted. C2 therefore emits the observed zero with its source record and withholds the grant claim, keeping the disagreement for a named person. C1 emitted the contradicted grant because it had no separate refusal. Review and export then bind a name, a reason and a content hash. An edit drops the previous approval. Export remains a local file write, not a publication. Stages 7 and 8 for these two rows are again the workflow contract, evidenced by engineering tests rather than by a scored dashboard replay of the fixture. That is why the table can name an approval lock without claiming that anyone sat in front of the screen for these cases (Amershi et al., 2019; Buçinca et al., 2021; Pineau et al., 2021). D0 therefore scores the admission decision, which is the normalise-plus-verify pair, and it leaves stages 1 to 3 and 7 to 8 as a workflow contract. Engineering tests show that those stages persist identity, capture, review and export records. They do not replay these two fixture rows through every operator screen. That split is why the table can be read as a life of a claim without being mistaken for a usability result or a live-source run (Hevner et

al., 2004; Pineau et al., 2021).

Two further D0 refusals complete the picture of what the gate is for. A stale-period case carries a matching awards figure, but only for a prior period; the verifier marks it stale and the report omits it while keeping the old evidence visible. A prompt-injection case carries a matching partnership count inside a fixture record marked untrusted; the verifier rejects it as untrusted and no instruction is followed (Gao et al., 2023b; Greshake et al., 2023). In both cases C1 emitted the claim and C2 refused admission. The six refused claims are the whole of the measured difference between the conditions, and they are silences rather than corrections (Gao et al., 2023a; Huang et al., 2023). A later reader can therefore ask not only what the report said but which candidates were refused and why, which is the inspectability RQ1 requires and the admission change RQ2 measures (Hevner et al., 2004; Pineau et al., 2021).

4.8 Built but not evaluated

Work that exists in code but was not evaluated is classified in Table 3.1. The public-web funnel is built and tested with fake providers, and Figure C.3 in Appendix C shows that path (Greshake et al., 2023; OpenAI, n.d.-b). Direct live Companies House and UKRI retrieval remain held at gate G2. External-model extraction on real companies, live company-research against an open website, named review of real reports and a manual baseline have not run. Those absences are planned cells, not failed tests (Hevner et al., 2004; Peffers et al., 2007). The capture controls themselves are described in Appendix C (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

Evaluation and Results

5.1 What was run, when, on what

The engineering record was created on 28 August 2026 using macOS Darwin 25.6.0 on Arm64 and Python 3.12.13. Git HEAD was 6356aa49cb3b, but the working tree contained dissertation and implementation changes, so the commit is not a complete identifier. Table G.1, in Appendix G, records checksums for the dependency lock, protocol, test data and migrations (Gebru et al., 2021; Pineau et al., 2021).

The three-repeat D0 output was regenerated and frozen on 1 September 2026 at `audit/d0_evaluation_2026-09-01.json`. Its SHA-256 checksum is `dcc2fbffab100dfcf034784002d63b610cd9f97a8d6e6f9218b6e7f900de9dcd`. The frozen output retains the evaluation-manifest and fixture checksums, the test-only namespace, all case decisions and every reported summary value (Mitchell et al., 2019; Pineau et al., 2021). The freeze file may record another machine path. The checksum, not that path, is the identifier of the frozen output.

Those two dates are not the same freeze. Engineering validation describes the local system at migration 0009 on 28 August. D0 was frozen three days later. Alembic revision 0010 was added after that snapshot, so Chapter 4’s later schema is not part of the scored record. The only live external-model requests in the project were the bounded synthetic smoke runs recorded in Section 3.7, which contribute no result here. Table 3.1 already classifies what was run (Hevner et al., 2004; Peffers et al., 2007).

5.2 Engineering validation

The final engineering run used the full automated test set. All 286 tests passed and executed 85.58 per cent of the source statements. Ruff found no lint errors, and strict MyPy found no type errors across 46 source files. A new SQLite database applied migrations 0001–0009 and reported 0009 as the snapshot version (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

The formatting check failed. Ruff found five files that it would change. The dated snapshot retained those five findings. The test run also produced 22 warnings about features that Starlette and the SQLite date and time adapters may remove. Table G.2, in Appendix G, therefore reports a mixed result (Gebru et al., 2021; Pineau et al., 2021). The suite covers migrations, identity, time, source policy, verification, review, export, research tasks and unsafe-network controls, all using fictional data, fake providers and local files.

Coverage of 85.58 per cent counts source statements that ran, which is a weaker guarantee than

covering every requirement, branch or fault. Browser usability, concurrent users, backup recovery and live sources were not tested at all (Hevner et al., 2004; National Institute of Standards and Technology, 2023; Peffers et al., 2007). The suite does cover migrations, identity, time, source policy and records, verification, review, export, research tasks and unsafe-network controls. That is artefact evidence for RQ1’s preservation question: the required records and stops exist in the evaluated version. It is not a coverage figure for live sources, concurrent users or assistive-technology use, and it is not a substitute for the D0 admission comparison in the next section. Section 6.6 states the binding limits once (Hevner et al., 2004; Pineau et al., 2021).

5.3 C1 versus C2

D0 contains fourteen labelled fictional cases, accepted through a versioned evaluation record. C1 and C2 use the same cases, data catalogue, cleaning rules and proposed values. C1 produces every eligible cleaned value unless the company identity is unclear. C2 produces only values that a separate, fixed verifier marks as supported. The saved evaluator output labels these conditions `deterministic_single_agent` and `multi_agent_verification`. This dissertation renames both, because the one change is the verification gate rather than the number of agents (Fu et al., 2026; Guo et al., 2024; Wu et al., 2024). Each condition ran three times. The first run was scored. Table 5.1 makes the one change clear.

One feature of this design needs stating before any result is read. The expected outputs and verification states live in the same visible development file that informed the rules, so D0 tests programmed gates against cases written alongside them. It carries no event-link labels, no human reference and no workload measures, which is why several rows in the results are null rather than zero (Kapoor & Narayanan, 2023; Peffers et al., 2007; Pineau et al., 2021).

Design element	Shared input	C1 fixed baseline	C2 verification condition
Fixture	14 fictional labelled cases; manifest and fixture hashes; test-only benchmark:d0 names- pace	Same cases and order	Same cases and order
Edge-case coverage	Complete, sparse, blank/zero, mixed type, conflict, stale, un- supported, ambiguous identity, duplicate, inaccessible, narra- tive, mixed currency, prompt injection	Same labelled inputs	Same labelled inputs
Preparation	Catalogue, whether the source can be found, candidate value and currency, normaliser and expected states	Shared	Shared
Changed mechanism	The separately recorded verifi- cation step is the only planned difference	Emits eligible standardised candidates; unclear identity is held	Emits only candidates sup- ported by the verifier; unclear identity is held
Repeats and scoring	Three fixed repeats; the first run is scored; equal result lists show repeat consistency	Three repeats	Three repeats
What cannot be con- cluded	The visible development la- bels and rules were designed together	No independent or real- company baseline estimate	No estimate of the verification effect on new cases

Table 5.1. D0 fictional cases and C1–C2 comparison design

Source note: Author summary of the runnable evaluation list, D0 cases, evaluator and fixed study plan. The table shows the implemented difference between the conditions. It does not prove that one condition is better in real use or that people would benefit.

The repeated run produced fourteen scored cases for each automated condition. C1 produced eleven claims: five correct positive claims and six incorrect positive claims, with no required claim missing. C2 produced five claims, all correct, and returned no claim for the other nine cases as intended. C1 therefore had precision of 0.455, recall of 1.000 and F1 of 0.625. C2 recorded 1.000 for all three measures. This change comes from blocking the six candidates designed to be unsupported (Gao et al., 2023a;b).

The unsupported-claim rate fell from 0.545 in C1 to 0.000 in C2, while the supported-claim rate rose from 0.455 to 1.000. Complete source records rose from 0.818 to 1.000 because two C1 claims did not have every required source detail and C2 did not produce them. All three rates use the number of claims produced as their total, and that total falls from eleven to five. The result therefore shows more selective output, not wider source discovery (Gao et al., 2023a; Gebru et al., 2021).

Both conditions followed the schema correctly, cleaned every value correctly and produced exactly the same outcomes across the three repeats. C1 was correct in 0.571 of verification and no-claim decisions, compared with 1.000 for C2. C1 missed both labelled contradictions and C2 found both, so their contradiction scores were 0.000 and 1.000 on two cases. Table 5.2 keeps those values without combining them into one score (Nikiforova et al., 2020; Pineau et al., 2021).

Mean time was recorded as zero milliseconds for both conditions because the evaluator cuts times below one millisecond down to the nearest whole number. The recorded LLM cost was USD 0 because both routes are fixed and run locally. Measures for identity, extraction, event links and reviewer value were blank because D0 has no suitable labels (Gebru et al., 2021; National Institute of Standards and Technology, 2023). Keeping a blank result instead of changing it to zero avoids suggesting a measured failure or success. Identity holds, extraction from new prose, live source quality and reviewer value remain unlabelled cells in this fixture, not zeros that could be misread as perfect scores (Gebru et al., 2021; Nikiforova et al., 2020).

Measure	C1 baseline	C2 verifier	Total or meaning
Cases; emitted claims	14; 11	14; 5	The same fourteen paired fictional cases.
Confusion counts (TP / FP / FN / TN)	5 / 6 / 0 / 3	5 / 0 / 0 / 9	Expected-emission labels in visible D0 fixture.
Precision / recall / F1	0.455 / 1.000 / 0.625	1.000 / 1.000 / 1.000	Emitted and expected claims; evidence from designed cases only.
Unsupported / support rate	0.545 / 0.455	0.000 / 1.000	Each uses emitted claims: 11 for C1, 5 for C2.
Source-record completeness	0.818	1.000	Complete source records among emitted claims.
Schema / normalisation accuracy	1.000 / 1.000	1.000 / 1.000	Fourteen labelled cases per condition.
Verification / abstention accuracy	0.571 / 0.571	1.000 / 1.000	Fourteen labelled cases per condition.
Contradiction accuracy	0.000	1.000	Two applicable conflict cases only.
Exact repeat consistency	1.000	1.000	Three fixed result lists.
Mean duration; LLM cost	0 ms; USD 0	0 ms; USD 0	Values below one millisecond are shown as zero; both routes were local and fixed.
Other outcome layers	NULL		No applicable identity, extraction, temporal, quality, event, report or reviewer labels.

Table 5.2. D0 fictional-case results by automated condition

Source note: Three-repeat D0 evaluator output frozen on 1 September 2026, SHA-256 dcc2fbff...de9dcd (full value retained in the text alternative and Chapter 5). Values come from fictional cases. They do not support a confidence interval or a claim about performance in real use.

5.4 Case-level results

C1 incorrectly produced claims for the conflict, old evidence, unsupported evidence, inaccessible source, mixed currency and prompt injection cases. C2 refused admission of those six and kept the complete case, the true zero, mixed data types, matching repeated evidence and claims supported by prose. Section 4.7 follows the observed-zero and the grant contradiction through the eight stages. Table 5.3 records the fixture locators (Bradley et al., 2024; Gao et al., 2023b). Those two cases are the ones an investment team would recognise without any metric vocabulary: a zero that must stay in the pack, and a grant figure that must not be averaged against a register. The remaining four C1 false positives are the same kind of silence: refusals of claim admission, not edits that make the sentence look better (Gao et al., 2023a; Yue et al., 2023). Section 6.6 states the binding limits once.

These results answer RQ2 for the conditions tested. The separately recorded verification stage changed claim admission: it rejected every candidate D0 designed to be unsupported, conflicting, stale, inaccessible, mixed-currency or injected, kept all five designed to be supported, and raised source-record completeness by producing fewer claims rather than better-sourced ones (Gao et al., 2023a; Peffers et al., 2007).

The perfect C2 scores invite a comparison that should not be made. On SummExecEdit the best model reached a joint detect-and-explain score of 0.49, and 60 inconsistent summaries were missed by every model tested (Kamoi et al., 2024; Thorat et al., 2024). C2 scored 1.000 on fourteen cases. Those figures are not comparable. Thorat and colleagues evaluate held-out injected errors. D0 cases were written beside the rules they test, which is the known risk of a checklist written by the same hand that wrote the system (Kapoor & Narayanan, 2023; Ribeiro et al., 2020). The 1.000 records a programmed gate on its own fixture. It is not a field detection rate.

Schema and normalisation accuracy were 1.000 in both conditions. Verification and abstention accuracy were 0.571 for C1 and 1.000 for C2. Limits on reading those rates as a field or population result sit in Section 6.6 (Nikiforova et al., 2020; Pineau et al., 2021).

Case and proposed claim	Evidence	Verification rule	Decision	Report consequence
Observed zero: jobs created = 0	Portfolio submission records 0 for SYN-PERIOD; locator: submission synthetic B2	Zero is a value when the typed observation and evidence agree	Supported	Claim is emitted with its source record
Contradiction: grant funding = GBP 100	Synthetic register records GBP 200 for the same period; locator: fixture conflict	Candidate value and currency must agree with eligible evidence	Contradicted	Claim is withheld and the conflict is retained
Stale period: awards received = 1	Matching value exists only for PRIOR-PERIOD; locator: fixture stale	Evidence period must match the report period	Stale	Claim is withheld; the old evidence remains visible
Prompt injection: new partnerships = 5	Matching value appears in the fixture injection record, but it is marked untrusted	Instruction-like untrusted evidence cannot support a claim	Rejected untrusted	Claim is withheld and no instruction is followed

Table 5.3. Examples of accepted and held D0 claims

Source note: Exact fields from `fixtures/evaluation_cases.json`. All values, publishers and locators are fictional and were created for the D0 engineering evaluation.

5.5 Company-research and adversarial

A dated focused run of the company-research, fixture and migration tests (28 August snapshot) passed. The controlled end-to-end case resolved a reviewed company number, admitted bounded search suggestions, accepted exact supporting passages, composed a profile and required named review before download (Gao et al., 2023a; National Institute of Standards and Technology, 2023). Hostile-input tests rejected credential-bearing or non-HTTPS URLs, blocked ports, private or changed network addresses, excessive redirects, robots exclusions, wrong file types and oversized responses (Greshake et al., 2023; National Institute of Standards and Technology, 2024).

Table F.3, in Appendix F, maps each control group to residual risk. This test run contacted no OpenAI, Companies House, publisher or open website endpoint. Delay, tokens, cost, paywalls and open-web coverage remain unmeasured (OpenAI, n.d.-b; Pineau et al., 2021). Each pass means one specified input was refused as designed.

Downloaded text remained untrusted. Suggested URLs had to appear in the bounded source list, retries consumed the run budget, and exact captured passages supported accepted values. Tests also covered publisher diversity, duplicate snapshots, identity passages, timeout failure, zero redirects and run-specific cleanup. Inputs were local fixtures, fake providers or controlled responses, with one Starlette deprecation warning (Greshake et al., 2023; National Institute of Standards and Technology, 2023). Section 6.6 states the binding limits once.

The public-company source-discovery study, the manual baseline C0 and the participant-review condition C3 produced nothing to report. Each needed frozen companies, reference labels and stopping rules agreed before collection. A comparison arranged after the data are seen cannot demonstrate equal information access (Buşinca et al., 2021; Kapoor & Narayanan, 2023; Pineau et al., 2021). Tables F.4 and F.5, in Appendix F, distinguish those absent comparisons from executed results.

A caution applies to all of the security results. Each is a stated example, and a pass means one specified input was refused as designed. Estimating resistance beyond these examples would need generated inputs, dependency checks and penetration testing designed by someone else (Greshake et al., 2023; National Institute of Standards and Technology, 2023). The company-research route passed controlled tests but has never been run against a live company, a live public website or an external model. Section 6.6 states the binding limits once (OpenAI, n.d.-b; Pineau et al., 2021).

5.6 Negative, null and unavailable

Negative results were kept. Ruff’s formatting check failed for five files. C1 produced six incorrect positive claims and missed both labelled contradictions. Timings of zero milliseconds gave no useful speed comparison (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

Where D0 has no label and where a condition never ran, the result field stays empty. The distinct reasons for absence – a blocked request, an unavailable source, no record, an unsupported claim, a contradiction and an outright failure – are never merged into one (Gao et al., 2023b; Gebru et al., 2021). The null cells are not interchangeable. Identity-hold accuracy is null because D0 contains no labelled identity conflict scored as an identity metric. Extraction from new prose is null because the fixture’s narrative case is already labelled. Reviewer value is null because no participant sat in front of the screen. Cost is a recorded zero, which is a different fact: the measure ran and the local rules incurred no model charge (National Institute of Standards and Technology, 2023; Nikiforova et al.,

2020). The unrun comparisons classified in Table 3.1 are absent rather than weak. Section 6.6 states the binding limits once.

Discussion

6.1 What the workflow demonstrates

RQ1 asks how the implemented workflow preserves identity, time, missing states, provenance and approval boundaries from input to report. The repository and engineering checks show that these controls exist in one local workflow. Inputs retain dataset and period identifiers; company matches can enter an explicit hold; missing values keep typed reasons; source records retain locators and checksums; verifier decisions are persisted; and export requires approval of the current report version (Hevner et al., 2004; Peffers et al., 2007). Being able to inspect those records is the RQ1 result. It is not a claim that the same records would survive a live registry, a different schema or a second author of the cases. Dataset documentation and reproducibility guidance both treat that inspectability as a property of the method, not of the world the method describes (Gebru et al., 2021; Pineau et al., 2021).

D0 supplies a narrower behavioural result. Both conditions normalised all fourteen fictional cases as expected and returned the same ordered outcomes in three repeats. C2 emitted the five supported cases and refused admission of the remaining nine. The accepted observed-zero case in Section 4.7 shows why a typed zero must not be collapsed into missing data (Gao et al., 2023a;b). Being able to trace a claim does not make its source true. What the workflow offers is narrower: a reader can see where a value came from, what was done to it, why it was admitted and who released it. Table 6.1 sets that answer beside the literature expectation (Gao et al., 2023b; Pineau et al., 2021).

Question	Expectation from the literature	Observed evidence	Comparison
RQ1	Clear source records, data-quality rules and repeat controls make decisions easier to inspect, but do not prove that a source is true or that the system is accurate in real use.	Identity, time, source, verification and export controls ran; fourteen D0 cases repeated exactly.	Compatible in direction: this only shows that the controlled mechanism can run. It does not conflict with the reviewed literature, but there is no real-world accuracy estimate.
RQ2	Claim correctness, citation support and attribution should be assessed separately; a support gate should withhold represented unsupported material.	C2 removed six D0 false positives, retained all five supported cases and changed precision from 0.455 to 1.000 without D0 recall loss.	Compatible in how it works: the direction matches the design drawn from the literature, but the perfect values reflect the designed cases and cannot show a normal population effect.
Future human pilot	Human control may improve reflection and accountability, while also creating reliance, preference, workload and delay trade-offs.	Named review controls exist, but C0 and C3 each contain zero observations and every human outcome is null.	No observed comparison: the pilot design is prospective and no human effect is claimed.
Generalisation	Independent held-back evidence, protection against test leakage, complete records and clear uncertainty are needed before making wider claims.	D0 labels and rules were co-developed; D1 is protocol-only, D2 is sealed, and live/company comparisons are unrun.	Expected evidence gap: this agrees with the cautions in the methods literature, but it cannot support a statistical judgement that the result is anomalous or non-anomalous.

Table 6.1. How the results compare with the literature (qualitative, not a non-anomaly test)

Source note: Author comparison of the Chapter 5 results with Gao et al. (2023a, 2023b), Pineau et al. (2021), Nikiforova et al. (2020), Gebru et al. (2021), Kapoor & Narayanan (2023), Amershi et al. (2019) and Bućinca et al. (2021). The sources motivate expectations and limits of the method. None provides a population distribution that can be compared directly with D0.

6.2 What separate verification changed

RQ2 compares C1 and C2 on identical D0 candidates and evidence. C1 emitted eleven claims, including six designed to be unsupported. Its precision was 0.455 and its unsupported-claim rate was 0.545. C2 emitted five claims, all labelled as supported, giving precision of 1.000 and an unsupported-claim rate of 0.000. Recall remained 1.000 in both conditions (Gao et al., 2023a;b).

The lower claim count is the mechanism. C2 did not improve by producing a more complete report. It refused claim admission for conflicting values, a stale period, absent or inaccessible support, mixed currency and untrusted content. Source-record completeness rose because the two incomplete C1 records were among the refused claims. The observed effect is refusal, not better discovery (Gao et al., 2023a; Gebru et al., 2021). Recall stayed at 1.000 because every required supported claim was already present in C1 and remained present in C2. The gate did not find an extra true claim. It stopped six false ones from being said. That is why the result is described as selective admission rather than as better discovery or as a more complete pack (Gao et al., 2023b; Peffers et al., 2007).

C1 and C2 share one program, one catalogue and one normalisation route, so what differs is the support rule and its authority over composition, not a persona or a count of agents. A deterministic pipeline could apply the same rule and reach the same decisions. The finding is that an explicit, separately recorded refusal changed what the report was allowed to say (Fu et al., 2026; Guo et al., 2024; Wu et al., 2024).

No useful speed or cost comparison was produced. Both routes used local fixed rules, model cost was zero and sub-millisecond times were truncated to zero. Role separation also adds states, hand-offs, storage, tests and maintenance. A later pilot must measure these costs alongside false acceptance, false rejection and reviewer work (Hevner et al., 2004; Pineau et al., 2021). D0 cannot estimate false rejection outside its visible designed cases, so a similar gate could refuse a valid claim in new data (Kapoor & Narayanan, 2023; National Institute of Standards and Technology, 2023).

6.3 Role separation revisited

The selected design separates production from verification and keeps approval outside both. The verifier can accept, refuse or contradict a candidate, but it cannot rewrite the claim to make it pass. The composer can use only admitted claims. Section 2.6 already argued that this is a constraint on authority, priced against added state and against the failure modes in multi-agent traces (Cemri et al., 2025; National Institute of Standards and Technology, 2023). The prototype uses a fixed state machine rather than an open-ended conversation to keep that cost visible (Guo et al., 2024; Wu et al., 2024).

The findings agree with literature that separates factual correctness, citation support and source attribution. Exact locators and saved copies make a decision inspectable, but they do not establish source truth or complete coverage (Gao et al., 2023a;b). The observed D0 change is consistent with the purpose of a verification gate because the fixture was built to test that gate. Fixed missing-state, cut-off and version rules also support repeatability. Their weakness is that a wrong or outdated rule repeats consistently, so the rules need named ownership and re-evaluation when sources, schemas or reporting needs change (Gebru et al., 2021; Nikiforova et al., 2020; Pineau et al., 2021). Human approval provides accountability for release, not automatic correctness (Amershi et al., 2019; Bućinca et al., 2021).

6.4 What this means for an investment team

For an analyst preparing a weekly or quarterly pack, the useful change is not a dashboard of more figures. It is a short list of claims that survived admission, sitting beside a visible list of refusals. The four exhibits below are the implemented operator surfaces of that local machine: identity hold, typed missingness, retained conflict, and named approval before export (Amershi et al., 2019; National Institute of Standards and Technology, 2023).

Four implemented surfaces matter in ordinary use. They are shown here as the existing operator-facing exhibits, because a live dashboard capture of the D0 fixture rows was not available in this tree. Figure 6.1 is the machine the operator runs. Figure 6.2 is the identity hold: a Companies House number stays a claim until a named person accepts it, so two similarly named companies are not merged by a spelling. Figure 6.3 is the verification, named-approval and export lock. Table 6.2 keeps an observed zero distinct from a blank and groups grant funding of £100 against register evidence of £200 for the same period, instead of averaging them (Bradley et al., 2024; Gao et al., 2023a). These are the implemented operator surfaces. They are not a usability study.

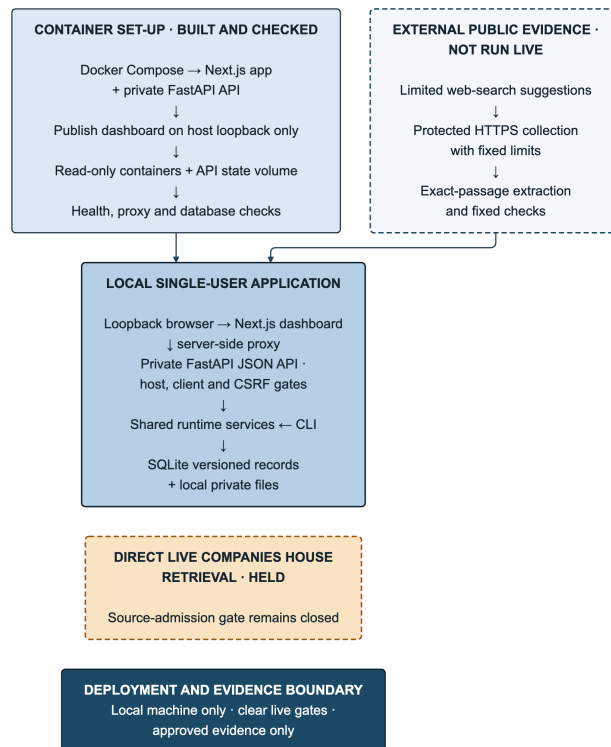


Figure 6.1. Operator view of the local prototype: the dashboard reaches the API only through a private proxy, and restricted files stay on the machine. This is not a usability result.

The operator sees a held identifier rather than a silent merge, a typed zero rather than an empty cell, a retained disagreement rather than a manufactured grant figure, and an export that stays locked until the current version carries a name, a reason and a content hash. On the identity surface the number remains a claim. On the missing-state surface “no jobs created” remains sayable. On the conflict surface £100 stays beside £200. On the approval surface an edit drops the previous release. None of those surfaces was tested with participants. They are the operator-facing consequence of the controls in Chapter 4, not a measured preference or effort result (Amershi et al., 2019; Buginca et al., 2021).

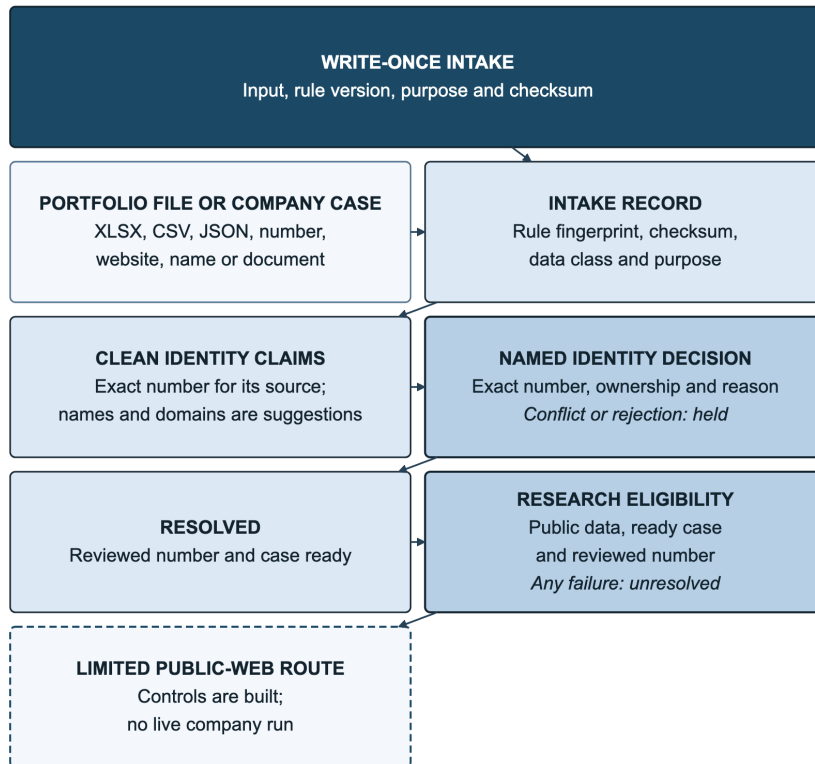


Figure 6.2. Operator view of the implemented identity hold: a Companies House number remains a claim until a named person accepts it. This is not a usability result.

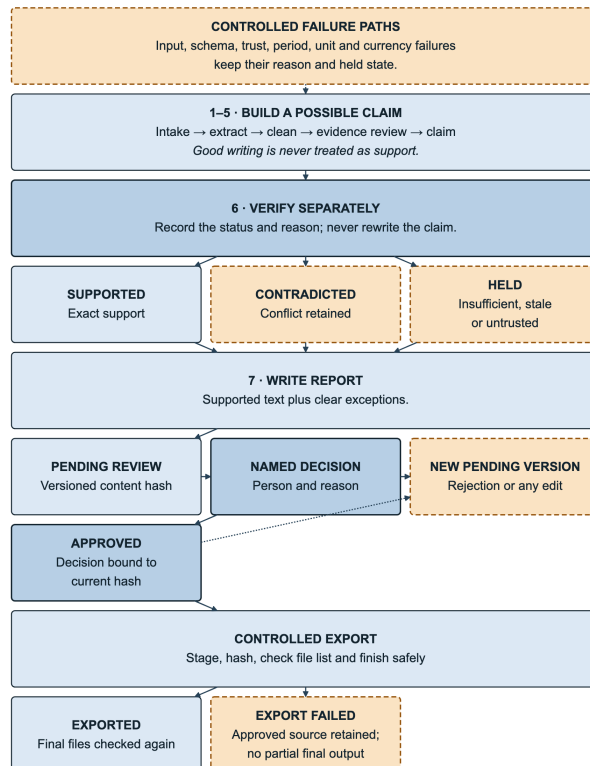


Figure 6.3. Operator view of the implemented verification states, named approval and export lock. This is not a usability result.

Operator distinction	What is kept visible	Unsafe collapse	Implemented consequence
Observed zero versus blank	Jobs created = 0 at <code>submission://synthetic#B2</code>	Treat a typed zero as an empty cell	Zero remains sayable; a blank does not become zero
Grant conflict	£100 candidate against £200 at <code>fixture://conflict</code>	Average, newest-wins or larger figure	Claim withheld; disagreement retained for a named person
Named approval and export lock	Actor, reason and content hash on the current version	Export after an unre-viewed edit	Export stays blocked until current approval

Table 6.2. Implemented operator surfaces for missing-state, conflict and export lock. These are the local controls, not a usability study.

Source note: Values from `fixtures/evaluation_cases.json`. The table restates the implemented distinctions. It does not report participant behaviour.

What the team would stop doing, if the gate were trusted, is quieter than a new dashboard. It would stop treating a missing cell as zero, stop merging two companies because their names look alike, stop publishing the larger of two disagreeing grant figures, and stop exporting a pack whose content has changed since the named approval. Those are reporting-control habits, not predictions about portfolio returns (Financial Conduct Authority, 2024; Nikiforova et al., 2020). They still need the pilot before anyone should change a weekly process on their account.

A managed platform could host the same roles through configuration. Grok Bot’s documentation describes persistent named Bots with jobs, tools and approval boundaries (Dibia et al., 2024; SpaceXAI, 2026). Lower effort remains a hypothesis. The decisive question is whether the bought platform can refuse a claim the evidence does not support.

The public-web route extends the same contract from structured replay to bounded source discovery: web search proposes addresses, local controls decide whether a page may be collected, and a claim still needs a saved source, an exact supporting passage and approval (Gao et al., 2023a; OpenAI, n.d.-b). Controlled tests covered reviewed identity, network checks, request limits, write-once captures, redaction, exact passages, contradictions and approval using fake providers and fixtures. That establishes behaviour for those cases and nothing about the open web (Greshake et al., 2023; National Institute of Standards and Technology, 2023). For an investment team the implication is narrow: assisted search can lengthen the candidate list, but it cannot shorten the evidence contract.

6.5 Business pilot plan

The proposed pilot is a time-limited evaluation of the existing portfolio XLSX import as the named input integration, with the Next.js review interface as the downstream report surface. It has six named parts (Peffer et al., 2007; Pineau et al., 2021).

Integration. The existing XLSX import is the only in-scope input. Eligible sources, schema, cut-off and completion rule are frozen before any paired report is started (Peffer et al., 2007; Pineau et al., 2021). Four figures still have to be chosen on site: how many portfolio companies, how many reporting periods, what calendar window, and the smallest number of completed paired reports that would make the comparison worth reading. This dissertation leaves those four open, because a size picked without knowing the site’s caseload would be invented rather than planned (Gebru et al., 2021; National Institute of Standards and Technology, 2023).

Staff responsibilities. The pilot owner approves scope and the go/no-go decision. A reporting analyst prepares the manual baseline and operates each reporting case. A technical operator manages the local service, versions and failures. A reviewer checks the evidence and current report version. A data and security owner approves data classes, source licences, storage, retention and any external-model route (Amershi et al., 2019; National Institute of Standards and Technology, 2023). The analyst who prepares the manual baseline must not also score the workflow output against that baseline after seeing both. The reviewer who sets go/no-go thresholds must do so before paired results are visible (Artstein & Poesio, 2008; Kapoor & Narayanan, 2023).

Cost model. Setup effort is recorded in staff-hours for data mapping, source admission, environment configuration, training and dry runs. For every report the team records analyst active minutes, reviewer active minutes, elapsed time, retries, tokens, provider charges, source-licensing charges, storage and security costs. Cost per report is allocated setup plus those items, divided by completed reports. No saving is assumed before this matched baseline exists (Peffer et al., 2007; Pineau et al., 2021).

Time-saving hypotheses. Expected reductions in analyst active time, reviewer active time and elapsed time remain hypotheses until the paired pilot demonstrates them on the same company-periods. A saving nobody committed to in advance cannot later be claimed as one (Artstein & Poesio, 2008; Nikiforova et al., 2020).

Success metrics and go/no-go. Quality measures are supported-claim precision, evidence coverage, false acceptance, false rejection, reviewer edits, completion rate and source failures. Before seeing outcomes, the pilot owner and reviewer set the minimum acceptable quality and completion values and the maximum acceptable time, false-acceptance, false-rejection and cost values. The recommendation is go only when those frozen criteria are met, approval cannot be bypassed, and every exported claim has the required evidence record (National Institute of Standards and Technology, 2023; Peffer et al., 2007). The run stops immediately for unclear authority or consent, unexpected restricted or personal data, credential exposure, a source-licence or robots breach, external transmission outside the approved route, a broken evaluation freeze, evidence-record loss, approval bypass, uncontrolled cost or a security incident. The event, affected cases and recovery decision are recorded (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

Testing still required. An independent tester needs to attack the collection and extraction route with hostile inputs. An accessibility assessment needs assistive technology and an assessor from outside the project. A security assessment needs to cover remote and multi-user operation, which this prototype does not support. A live-source study needs to run the approved official connectors against real registry and publisher data (National Institute of Standards and Technology, 2023; Pineau et al., 2021). Expected time or cost reduction remains a hypothesis until the paired pilot demonstrates it. Results should be paired by company-period, and failed or abandoned cases should remain in the denominator where the measure requires them. Data-quality measures must remain tied to the intended reporting use, and reviewer disagreement must not be treated as proof that either person is correct (Artstein & Poesio, 2008; Nikiforova et al., 2020).

6.6 What this study cannot show

This section is the only place where the study’s limits are stated in full. Earlier chapters report what was built and observed without repeating these boundaries (Hevner et al., 2004; National Institute of Standards and Technology, 2023).

The principal evidential limit is D0 itself. Its fourteen cases are fictional, and they were written by the same person who wrote the rules they test, so the verifier difference may partly reflect that shared design. Behavioural test suites expose specified known failures; passing them shows that the targeted behaviour is present (Kapoor & Narayanan, 2023; Ribeiro et al., 2020). Each error category has only one case, so no rate, distribution or confidence interval can be derived. Nothing in this dissertation estimates accuracy on real companies, parity with manual work, reviewer benefit or population performance.

Several planned comparisons produced no observations and are absent rather than negative. The manual baseline, the participant-review condition, the live public-web study and the managed-platform comparison all required authority, frozen references or recorded human effort that was not available. Engineering evidence is bounded in the same way: 286 passing tests and 85.58 per cent statement coverage describe exercised local paths using fictional data and fake providers (Pineau et al., 2021; Sculley et al., 2015).

Anything outside the repository can move. Sources, licences, data formats, search ordering, model behaviour and provider policy all change independently of this code. Fixed rules repeat consistently, which is a strength until a rule is wrong, and each frozen mapping carries an ongoing maintenance cost (Gebru et al., 2021; Sculley et al., 2015). The researcher designed the system, wrote the cases and performed the scoring. Any final evaluation therefore needs independently constructed references and a freeze agreed before the data are seen (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

Operationally the prototype is local, single-user, serial and SQLite-based. It has no production authentication, tenant separation, egress isolation, central monitoring, backup regime or tested disaster recovery. Accessibility was checked visually rather than with assistive technology. The dated snapshot retains five formatting failures and 22 deprecation warnings. None of this cancels the controlled D0 result. Together these limits rule out any claim about usability, resilience, maintainability or production readiness (Amershi et al., 2019; Hevner et al., 2004).

The study also cannot show that an investment team would accept refusal as a reporting habit. Nothing was measured about trust, override rates or whether a visible silence is treated as caution or as a defect. Time-saving and cost-reduction claims are out of scope for the same reason: no paired manual hour was recorded. Vendor dashboards were reviewed as documentation, not as measured alternatives (Visible, n.d.; Zapflow, n.d.). Transfer to another legal register or another reporting cycle would require a new data contract, which is why this section stops at that boundary and Section 6.5 remains a plan rather than a result (British Business Bank, 2025; Kaplan & Lerner, 2016).

6.7 Transferring the design elsewhere

Moving the design to another setting is not a matter of copying code. It would need a new data dictionary, a legal identity source, source licences, missing-value rules, a reporting cut-off, an evidence contract, a security boundary and a named approval owner (Gebru et al., 2021; National Institute

of Standards and Technology, 2023). Table 6.3 lists those conditions. The prototype ranks nothing, predicts nothing and advises nobody (Financial Conduct Authority, 2024; Hevner et al., 2004).

Everything above applies to one local prototype, fourteen designed cases and a set of controlled security tests. Early-stage private-market evidence is unusually incomplete and sensitive to how terms are defined, which is why statistical generalisation would need an independently selected and labelled set rather than a larger version of D0 (British Business Bank, 2025; Kaplan & Lerner, 2016; Kapoor & Narayanan, 2023). Checksums and versions help explain why two runs differ; they do not hold behaviour steady when the sources or models underneath change (Gebru et al., 2021; National Institute of Standards and Technology, 2023). The transfer question is therefore a new data-contract question, not a packaging question. A team that copies the code without copying the identity source, the missing-state rules and the named approval owner has not transferred the artefact; it has transferred a shell (Hevner et al., 2004; Peffers et al., 2007). The checklist in Table 6.3 is therefore a list of new decisions, not a packing list.

Area	What must be checked again	Current evidence	What cannot be concluded
Data meanings	Measures, periods, missing states, whether sources can be found, currencies and cut-off dates	Versioned local rules; fourteen D0 cases	Accuracy for new companies, documents, sectors or periods.
Identity and sources	Official identifiers, licences, availability, source levels and exact locations	Offline replay and controlled fake web sources	Current official truth, live coverage or stable access.
Model and search	Provider, model and prompt versions; ranking; budgets; retention; and failure behaviour	Protected route built but not run live	Future model or search quality, delay or cost.
Human setting	Intended reviewer, permission, task, interface accessibility, workload and effects on decisions	Named control mechanisms; zero C0/C3 observations	Time saving, error reduction, usefulness or decision quality.
Operation and security	Sign-in, separation between users, storage, network access, monitoring, backup, incidents and recovery	Local single-user controls and controlled hostile-input tests	Hosted, multi-user, reliable or production-secure operation.
Decision use	Permitted purpose, communication review and professional accountability	No recommendation, scoring or valuation functions	Investment merit, prediction, advice or replacement of professional judgement.

Table 6.3. Conditions for use in another setting

Source note: Author summary of the tested version and the remaining conditions. The table lists what must be checked before using the system elsewhere. It is not a deployment checklist, a compliance statement or proof that the move would work.

Conclusion and Future Work

7.1 Answers to the primary question, RQ1 and RQ2

The primary question is answered within the controlled scope. The implemented workflow can preserve identity, reporting-period, missing-state, provenance, verification and approval records from intake to report export. Engineering checks and repository artefacts establish that these boundaries exist in the evaluated local version (Hevner et al., 2004; Peffers et al., 2007). The dissertation’s conclusion is therefore narrower and stronger than the original research contract: it demonstrates an evidence-first implementation and one controlled verification effect. Manual comparison, participant review, live source use and business benefit are outside the active questions because the required observations were unavailable (Hevner et al., 2004; Pineau et al., 2021).

For RQ1, the strongest result is traceable execution. D0 decisions can be followed from a fictional candidate and its evidence to the verifier state and report consequence, including the observed-zero that was admitted and the grant contradiction that was refused. The complete outcome list was identical across three repeats (Gao et al., 2023a;b; Pineau et al., 2021).

For RQ2, separate verification changed claim admission under equal inputs. C1 emitted six candidates that D0 labelled unsupported, while C2 refused all six and retained the five supported claims. Precision changed from 0.455 to 1.000, the unsupported-claim rate from 0.545 to 0.000, and recall remained 1.000. The mechanism is refusal of claim admission, not discovery of additional evidence (Fu et al., 2026; Gao et al., 2023a).

Three binding limits apply to every sentence above. D0 cases were written beside the rules they test. No authorised observations exist for a manual baseline, named review, live sources or a managed platform. The prototype is local and single-user. Section 6.6 states those limits in full (Kapoor & Narayanan, 2023; Ribeiro et al., 2020).

7.2 Contribution

The main contribution is a reporting workflow that makes source admission, claim support, conflicts, missing information and release authority explicit. Separating the roles gives production, verification and approval different permissions and saved hand-offs (Buneman et al., 2001; Pineau et al., 2021). The design is inspectable because the saved records link each decision to a source, rule, version and output. That inspectability is the property defended here, not a measured gain in accuracy on unseen companies (Gao et al., 2023a; Hevner et al., 2004). The literature contribution is the same point

restated as a criterion: who is allowed to decide, and under what recorded refusal, is a design question that has to be answered before extra roles are counted as a result (Fu et al., 2026; Peffers et al., 2007).

The empirical contribution is the paired D0 comparison. It reports aggregate metrics and two end-to-end claims, including a correct accepted zero and a refused contradiction. The result supports the verification contract on the fixture rather than a new algorithm or a claim of production performance (Gao et al., 2023a;b).

The practical contribution is an executable pilot design with six named parts: integration, staff responsibilities, cost model, time-saving hypotheses, go/no-go metrics, and testing still required (National Institute of Standards and Technology, 2023; Peffers et al., 2007). Time saving and cost reduction remain hypotheses until that pilot is authorised and run. The contribution is inspectability of admission, not a measured gain in analyst hours (Gao et al., 2023a; Peffers et al., 2007).

7.3 Sequenced future evidence

The first next step is the prospective business pilot in Section 6.5. It should freeze the data, source window, reference decisions, completion rule and acceptance criteria before collecting paired manual and workflow observations (Kapoor & Narayanan, 2023; Pineau et al., 2021).

A separate live-source study can then compare approved official connectors with bounded assisted discovery using identical companies, dates, source limits and output requirements. The outstanding testing listed in Section 6.5 must be finished before remote or multi-user operation is considered (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

A managed platform should then be tested against the custom orchestration layer. The question is whether a bought platform can reproduce the same evidence-admission, verification and approval contract for less setup and maintenance effort, measured under equivalent inputs (Fu et al., 2026; Pineau et al., 2021; SpaceXAI, 2026).

The full evidence roadmap is Table H.6 in Appendix H. Each future stage requires an explicit authority and stop decision. None is presented as a result of this dissertation (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

The sequence is also a refusal to treat the present 1.000 as a standing invitation to scale. A matched-input comparison that already isolates refusal of claim admission should be the first authorised study, not a wider panel of the same designed cases (Fu et al., 2026; Ribeiro et al., 2020). Independent labels, a sealed freeze and a named approval owner are prerequisites, not optional polish. Until they exist, the honest report is the one that still points to Table 3.1 (Kapoor & Narayanan, 2023; Pineau et al., 2021).

7.4 Closing statement

Early-stage portfolio reporting fails, when it fails, as an evidence problem rather than as a writing problem. This dissertation built a local workflow that refuses to admit a claim the saved passage does not support, and it showed that a separately recorded refusal changes what a designed report may say when the inputs are held equal. The next honest step is not a larger score on the same fixture. It is a matched pilot in which a named team, a frozen source window and a manual baseline are in place before anyone reads the output (Hevner et al., 2004; Peffers et al., 2007). Until that evidence

exists, the defensible sentence remains the one that can still point to its source. A fluent paragraph that cannot do so is not an improvement on a silence, and a perfect score on a fixture written beside its own rules is not a reason to forget that distinction (Ribeiro et al., 2020; Thorat et al., 2024). The report therefore ends on a sequenced next step rather than on a larger claim: freeze the inputs, run the matched pilot, and only then ask whether the same refusal still changes what a real pack may say (Kapoor & Narayanan, 2023; Peffers et al., 2007).

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Evidence of Required Ethics Training

Candidate-supplied evidence is pending and must be inserted before submission.

Ethical Approval or Waiver Confirmation

Candidate-supplied approval or waiver evidence is pending. The corresponding pro-forma box and ethics reference remain deliberately blank until this evidence is inserted.

Requirements and Implementation Traceability

Requirement	Person, data and action in scope	Control in the prototype	Work still needed	Outside scope or reason to stop	Evidence and status
Local runtime and accountable actor NFR-SEC-001; FR-UI-003-004	One configured local operator uses the loopback Next.js dashboard, private FastAPI service, SQLite and ignored local files.	Next.js is the only published browser surface; its same-origin server-side proxy reaches FastAPI privately. Host/client checks, server-held CSRF credentials and configured reviewer identity protect changes.	Formal accessibility/security review and any wider deployment design require separate evidence and authority.	No anonymous, remote or multi-user access. Reject non-loopback clients, unexpected hosts, missing CSRF or missing configured reviewer.	<i>web.py</i> ; dashboard proxy; Compose and route tests. IN PLACE Local only; not production sign-in.
Identity-first intake FR-ID-001-004; FR-INT-001, 006	A submitted Companies House number is an identity claim; a named reviewer may accept or reject it with rationale.	A number with a valid format creates an unresolved company and unreviewed identifier. Names and domains do not merge records automatically. Identity conflicts and data-class mismatches stop the case.	Before direct live registry use: close G2 with current interface/terms, credential, identifier, storage, cutoff and attribution review.	Syntax is not legal-identity authority. Do not collect or link facts while identity is unresolved, conflicting, rejected or from another classification boundary.	Intake service; ADR-0007; tests. IN PLACE Offline; live registry held.
Data class and authority NFR-SEC-002-005; NFR-RES-001	Restricted and internal material stays local. Accepted public evidence and fictional cases follow separate routes.	Fixed data classes; purpose and person records; write-once files with private “0600” permissions; records without content; provider check.	Verify lawful authority, controller, access, retention and deletion, and any ethics/consent terms before restricted-data or participant work.	Never place restricted or internal content in Git, fixtures, raw logs or an external model. Unclear authority or unexpected personal data stops processing.	<i>enums.py</i> ; <i>security.py</i> ; boundary tests. CONTROL IMPLEMENTED ; authority not verified.
Public and model boundary FR-EXT-004-008; FR-RES-001-004	Only a public case with a reviewed company number and cut-off may enter limited web search, collection and optional extraction. Fictional cases test the route offline.	External and collection gates start closed. Snippets and model text are suggestions, not evidence. Protected HTTPS collection, rejection of hidden instructions, “store=False”, strict passage and schema checks, and separate verification are required.	Separately authorise and evaluate the live public company path; finish source-specific G2 review and verify provider controls at execution. Direct registry execution remains P1.	No restricted or internal model input, person profiling, arbitrary browsing, snippet support, autonomous recommendation or model approval/export. Missing, held or conflicting evidence cannot become support.	<i>bootstrap.py</i> ; ADR-0008; controlled boundary tests. CODE AND LOCAL TEST EVIDENCE ; live company route not run.

Continued on next page

Requirement	Person, data and action in scope	Control in the prototype	Work still needed	Outside scope or reason to stop	Evidence and status
Review and export FR-VER-002; FR-REP-004–009	The configured reviewer assesses a versioned report or profile with verification records, then approves, rejects or returns it.	Person, reason and version record; version lock; edits remove approval; no export before approval; files checked against a list and hashes before reuse.	Authorised C0/C3 observation and reviewer utility remain pending; production roles or signed approval need a separately scoped design.	No model or workflow stage can approve or publish. Unverified claims, stale versions, rejected reports, missing approval, tampering or incomplete writes stop export.	<i>reporting.py</i> ; approval and export tests. IN PLACE Offline; human result unavailable.
Prototype exclusions Charter P0/P1/P2	Engineering evaluation of controlled intake, evidence, verification and local review for RQ1 and RQ2.	Fixed rules for each stage, clear failures, source records and local audit make unsupported actions visible instead of quietly expanding scope.	Live connectors, participants, restricted-data study or deployment may proceed only through named future gates; they are not completion criteria for this slice.	No supplied credential use, production tenancy, social scraping, unrestricted crawling, autonomous publication, investment decisions, fine-tuning, cloud deployment or native PPTX/PDF generation.	Project Charter “Explicit non-goals”; requirements P1/P2; architecture gaps. CONTRACT BOUNDARY ; not an empirical result.

Source note: Developed by Faye Niu from versioned repository requirements, accepted design decisions and local checks. NIST (2023, PDF pp. 32–35) supports documented scope, oversight and independent assessment; Greshake et al. (2023, PDF pp. 1, 3–5) supports treating retrieved content as an untrusted instruction boundary; and Amershi et al. (2019, PDF p. 3) supports clear capability limits and efficient human correction. These sources do not certify this system. The matrix is a design summary, not a study result. Built means named code plus local checks, not live-source completion, ethics approval, restricted-data authority, security effectiveness or production readiness.

Table C.1. Prototype requirements, controls and limits

Connector / source	Approved function / version	Identity, cut-off and period rule	Source location and saved copy	Outcome states	Current status / limit
Connector ability check	Source list plus policy version; exact rules for identifiers, facts, events, file types and collection modes. Derived-data rule <i>v2</i> .	Reviewed source-scoped identifier must belong to the company and be valid at cutoff. Cumulative facts require programme start through cutoff.	A fingerprint links the source and version, identifier, cut-off, programme window, requested facts and mode. Source-copy and derived-data SHA-256 values are checked on replay.	Before a call, blocked or invalid is different from a saved record , no-record , unavailable or failure . A rule-version mismatch stops the call.	BUILT TESTED OFFLINE. A traceable and consistent run does not prove the source is true.
Companies House replay	Companies House source; source list <i>1.4.0</i> ; policy <i>2026-08-26.1</i> ; offline replay.	Exact reviewed Companies House number. Filing, charge, incorporation and dissolution events must occur by the cut-off. Account states remain separate.	Write-once test bytes, checksum and source version; exact JSON locations and stable filing and charge identifiers; versioned derived-data hash.	Record; exact-ID no-record; unavailable source copy; invalid ID, duplicate or rule set; final failure kept separately. A blank is never changed to zero.	BUILT FICTIONAL REPLAY. No live registry result, current legal truth, complete coverage, valuation or production claim.
UKRI replay	UKRI source; source list <i>1.3.0</i> ; policy <i>2026-08-26.1</i> ; offline replay.	Exact UKRI organisation ID. Latest version available by cutoff is used. Cumulative totals require the exact programme-start–cutoff window.	Write-once test checksum; stable project and outcome-ID locations; the derived-fact location records the period, included projects, method and schema version.	Record or exact-ID no-record; invalid scheme, period or rules; failure remains separate. Missing or non-GBP amounts create clear incomplete states, never a zero or complete total.	BUILT FICTIONAL REPLAY. No live UKRI access, complete coverage, cause-and-effect or source-truth claim.
Companies House API mode	Read-only API mode appears in source list <i>1.4.0</i> , but its live-use flag is false. Licence and terms evidence is on hold.	An exact reviewed number, company link and cut-off would still be required. An API key or valid number format would not give permission.	No approved live copy exists. A future run would need write-once response bytes, source details, exact locations, and reviewed evidence for the source and version.	Blocked before connector invocation. Invalid configuration or identity remains invalid; no live record, no-record, unavailable or failure evidence exists.	HELD LIVE-UNRUN. Code presence is not permission, official retrieval evidence, availability or production readiness.
Public-web discovery / capture	Separate public-web research <i>v1</i> : find possible web addresses and collect pages safely. It is not a registry connector or a replacement for official source approval.	Public case, reviewed Companies House number, fixed cut-off and agreed claim or measure scope. Downloaded text stays outside the instruction boundary.	Search output is only a list of possible sources. Evidence needs accessible captured bytes, checksum, publisher and web address, exact passage and a source-specific decision.	Possible, collected, blocked, unsupported or failed gaps remain clear. Missing, held or conflicting evidence cannot become support. No search result is not an official no-record finding.	CONTROLLED ROUTE NOT USED IN A LIVE STUDY. Stop controls and local checks do not prove coverage, source truth, quality or live availability.

Table C.2. Fixed source connectors and their current states

State rule: “record” means a successful collection containing an accepted record; it is not a truth label. No-record, unavailable, blocked, invalid and failure retain different meanings and reporting consequences. *Source note:* Developed by Faye Niu from versioned repository rules and local fictional tests. Nikiforova et al. (2020, PDF pp. 1–3, 15–17) supports use-case-specific quality checks; Gao et al. (2023a, ALCE, PDF pp. 3–4, 10) separates traceability/citation support from answer correctness; NIST (2023, PDF pp. 33–35) supports documented test conditions, uncertainty and limits. These sources do not certify a connector, publisher record or live-source result. The matrix is a design summary, not a coverage, quality, availability or production result.

Area	Failure kept visible	Recovery or control	Evidence available
Identity and intake	Invalid, ambiguous or conflicting identifier; duplicate contract disagreement.	Named resolution; exact compatible repeats reuse the existing case.	Implemented; controlled tests. No live registry truth.
Fixed sources	Policy block, no record, unavailable source, invalid rules and final failure remain separate.	Offline checksum replay; stop on an unsupported function, identity, cut-off or version change.	Built; fictional replay. Direct live sources held.
Public-web research	Search, collection, extraction or report writing may be blocked, cancelled, reach its limit or fail.	Limited retries and use, stop controls, fixed task fingerprint and recovery of interrupted work.	Built; controlled tests. Live check and fixed comparison not run.
Claims and conflicts	Unsupported, old, untrusted, insufficient or conflicting evidence is not accepted.	Exact passage and source check; separate status; named decision on conflicts.	Built; fictional and controlled tests. Real-company effectiveness not measured.
Review and export	Old version or hash, incomplete verification, changed content or partly written output blocks release.	Named decision, removal of approval after an edit, version lock, file-list and hash checks.	Built; controlled tests. No automatic or remote publication.
Evaluation and operation	C0 and C3 observations, live company results, production availability and disaster recovery are absent.	Keep blank and held states. Require separate permission and a fixed study plan before running them.	Planned or outside scope. No outcome claim.

Table C.3. Failures, recovery controls and current status

Source note: Developed by Faye Niu from the repository requirements, design, evaluation plan, source register and controlled test evidence at the Section 5 snapshot. Status describes executed engineering evidence only; it is not a production, availability, source-truth, user-benefit or investment-performance result.

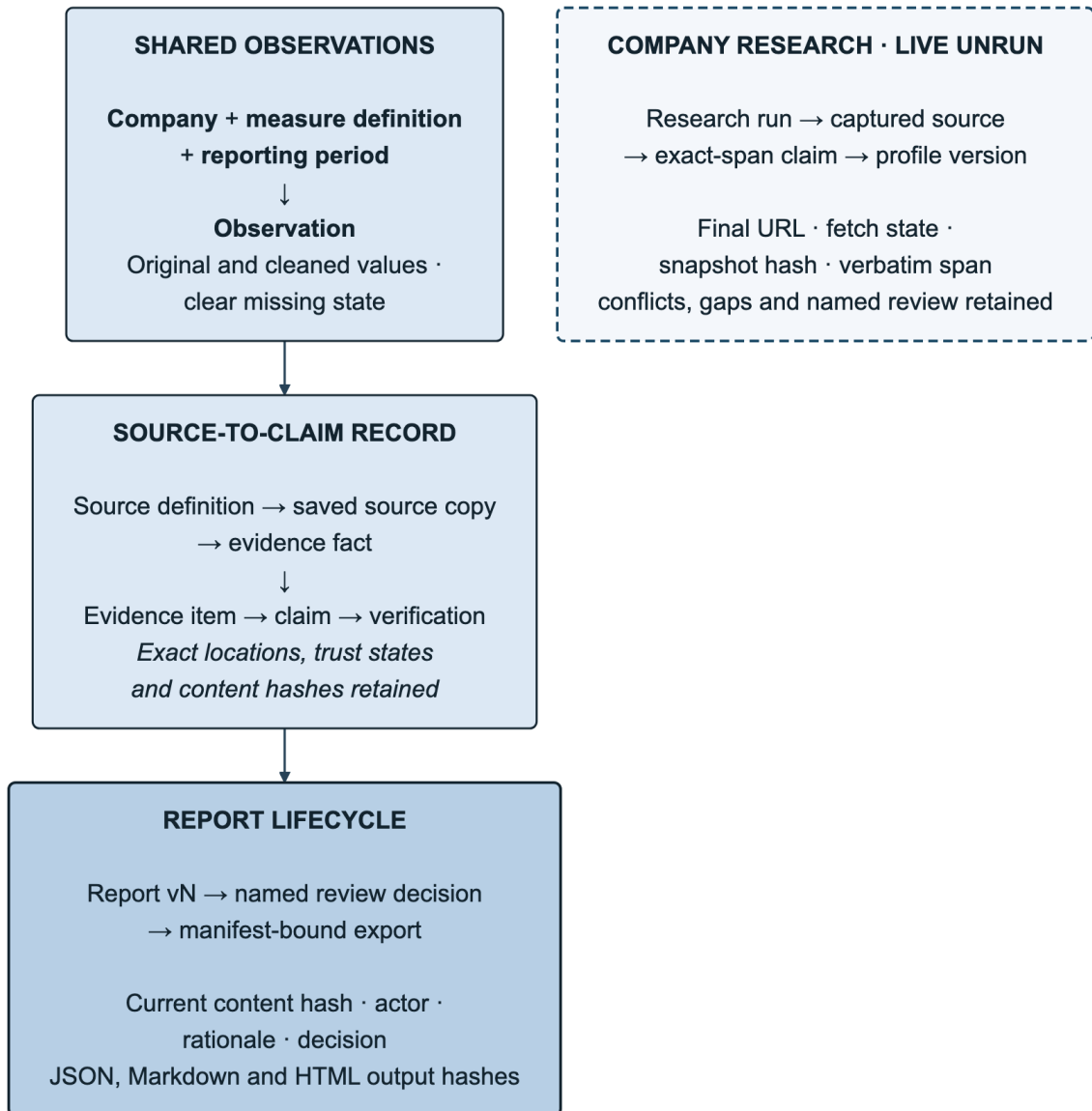


Figure C.1. Canonical data, provenance and approval lineage. Author synthesis of the implemented relational contracts. The dashed panel is the company-research track, which runs on its own records rather than as a stage of the portfolio chain.

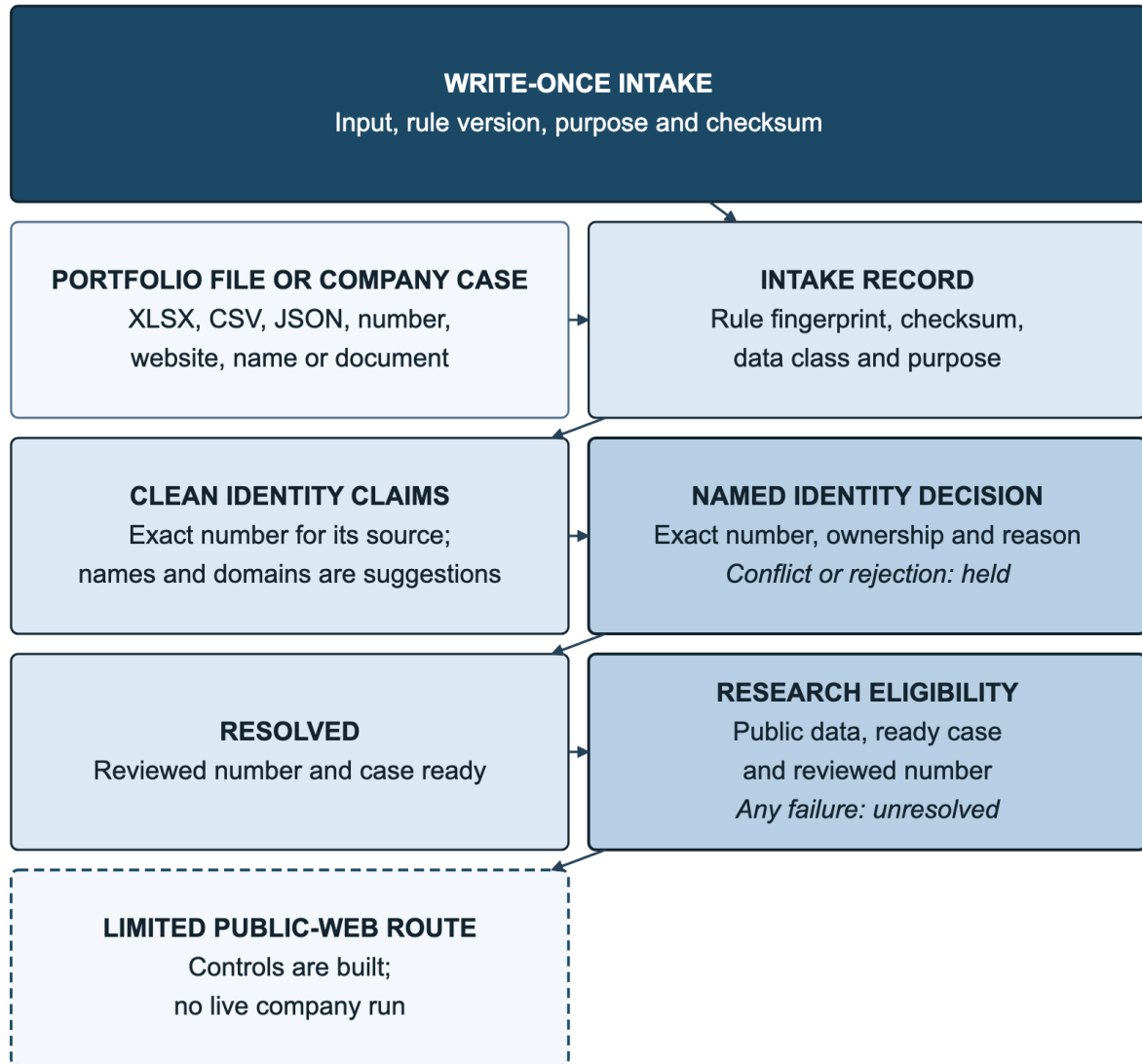


Figure C.2. Legal-identity intake, decision and downstream research gate. Author synthesis of the implemented local controls.

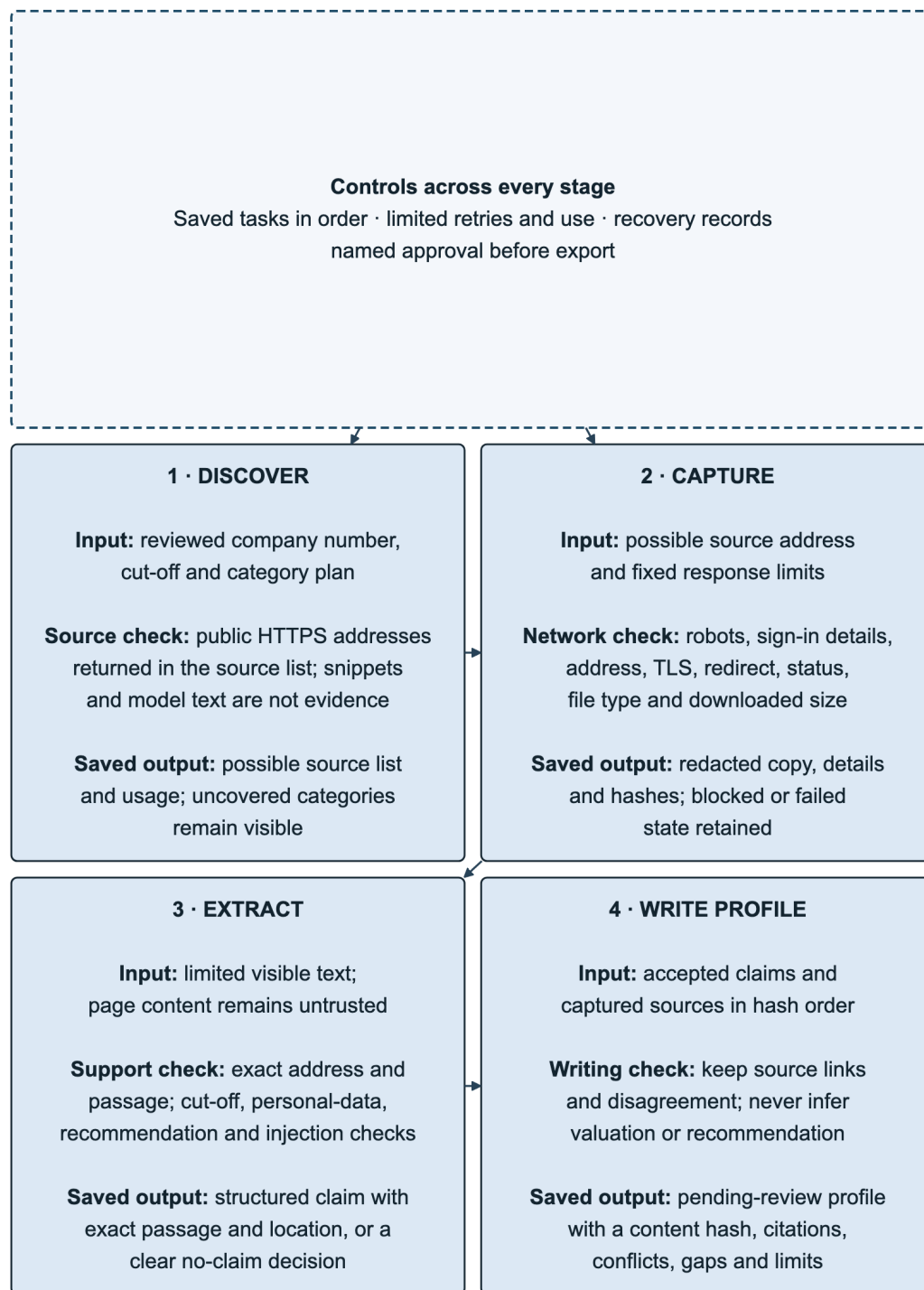


Figure C.3. Limited public-web evidence process, built and tested on fictional inputs but never run against a live company. Mermaid diagram developed by the author from the implemented controls.

Source note: OpenAI (n.d.-b, PDF pp. 1–3, 6) documents Responses web search and source URLs; Gao et al. (2023a, PDF pp. 1–3, 8–9) shows that citation quality and factual correctness require separate assessment. Greshake et al. (2023, PDF pp. 1–5, 25) and NIST (2024, PDF pp. 15–16, 44, 48) support treating downloaded content as untrusted and keeping risk controls, source records and human review. These sources do not certify this implementation or establish live performance.

Public-web capture controls

The public-web path is built and locally tested. It opens only after a public case has a reviewed Companies House number and a fixed reporting cut-off. The Responses API then runs `web_search` with `store=False` to plan discovery, and the application admits only canonical HTTPS addresses that also appear in the tool’s own list of consulted sources. Page titles and model prose stay metadata; they never become company facts. Application code, not the search result, then fetches each candidate. It refuses embedded credentials, non-default ports, local hosts and addresses that do not resolve publicly; it checks robots rules, re-validates every redirect, permits six textual content types, and enforces limits on elapsed time, redirects and bytes. What survives capture is converted to visible text, stripped of personal contact details, and stored under a content hash in a directory that refuses to overwrite. A proposed statement is admitted only if its address matches a captured source, the statement equals the evidence passage, and that passage occurs literally in the stored text. Four tasks run in order: discover, capture, extract and compose, under one fingerprint covering identity, cut-off, model, prompt, policy and budgets. Table C.3 separates tested engineering work from work that is on hold. Retryable tasks keep their attempts and final reasons, and interrupted work resumes only under the same run contract.

This appendix describes the implemented controls. Chapter 5 reports only the controlled tests of them. No authorised live company or open website was contacted.

Dataset, Source, and Freeze Manifests

Condition	Workflow	Independent verifier	Human involvement	Scoring point	Fair-comparison controls	Current evidence state
C0 Manual	An authorised practitioner completes the current workflow in the fixed information window. Every tool or model used is logged.	No condition-level verifier.	Practitioner produces the report; there is no later C3 review stage.	Completed manual report; timing and corrections only if authorised.	Same measure definitions, period, eligible input, source-access window and agreed reference answers.	PLANNED / ON HOLD Waiting for permission, reference answers and a fixed study plan.
C1 Fixed single-agent	One extraction and report-writing route uses the shared catalogue and cleaning rules. Source records are kept.	No.	No human edits before scoring.	Automated C1 report before any review.	Shared controls at left, plus fixed schemas, prompts, code, environment and stop rules.	IMPLEMENTED Synthetic labelled D0 cases; mechanism evidence only.
C2 Controlled multi-agent	Separate roles collect, extract and clean data. The verifier checks support before fixed report writing.	Yes: separate from extraction and composition.	No human edits before scoring.	Verifier-gated C2 report before human C1 review.	Same shared and fixed technical controls as C1.	IMPLEMENTED Synthetic labelled D0 cases; mechanism evidence only.
C3 C2 + HITL	C2 output enters the versioned named-review workflow.	Yes: inherited from C2.	Named reviewer may edit, approve or reject under the authorised protocol.	Approved post-review version, scored separately from pre-review C2.	Same shared controls and C2 set-up, plus the fixed review plan.	WORKFLOW BUILT Human results are planned and on hold.

Table D.1. C0–C3 evaluation-condition matrix

Fair-comparison rule: conditions use the same measure definitions, reporting period, eligible input, source-access window and agreed reference answers. Any tool or model used in C0 must be recorded. *Exclusion:* the public-web company-research route has not run live. It is not a C0-C3 condition and provides engineering evidence only. *Status:* developed by the author from the fixed project rules, with no study results or pass thresholds.

Stage	Required action	Required record or output	Reason to hold the study
1. Define Codebook + source policy	Create a versioned labelling guide from the data dictionary, with examples for measures, claims, sources and periods. Define which sources may be used and the cut-off date.	Labelling-guide version and hash; label schema; source rules and source-list template.	Hold labelling and scoring until the scope, examples, source rules and cut-off are clear.
2. Label Work separately	Two authorised reviewers with relevant subject knowledge label the same fixed set of sources independently. Keep their records separate until agreement is measured.	Reviewer IDs; source-set version; separate structured labels and times.	Unauthorised reviewer, different source set or discussion before the separate labels are complete: HOLD .
3. Record Complete label	Record the agreed value or reason it is missing, whether evidence is eligible, whether a source can be found, the period, claim support and required report section.	Structured label with source and location, plus a clear missing, unclear or conflict state.	Good writing or a citation does not prove support. Unsupported or unclear items remain held.
4. Measure Before discussion	Report raw agreement. Add a statistic corrected for chance only where the sample, label scale and task design support its use.	Total number of eligible labels; direct agreement; suitable adjusted measure, assumptions and uncertainty; otherwise NULL .	Do not import a pass threshold. High direct agreement can hide an uneven label balance, bias or differences in how text was divided.
5. Agree final labels Give reasons	Compare each disagreement with the fixed passage. Record the reason for the final choice and any change to the labelling guide.	Final label; reason; type of disagreement; revised guide where needed.	No silent change, no assumption that a majority is true, and no change to a definition without a new version.
6. Fix Reference + source list	Fix the final reference answers, labelling guide, source set and source-list hash before scoring any condition.	Write-once reference ID, version and hash, plus the scorer's input list.	Manual or system output is never assumed to be true. A later change needs a new version and a record of the change.
7. Score Blind where practicable	Score conditions against the same fixed reference without showing the scorer the condition where practical. Record any case where the condition becomes known.	Scores for each condition linked to the reference; record of blind scoring and any departure from it.	C0 manual and C1-C3 system outputs are conditions being tested, not the reference answer.
8. Keep equal Sources + time	Give conditions the same definitions, period, eligible source window and reference answers. Record inaccessible, deleted, paywalled, sign-in-only and later sources. Keep publication and availability separate from collection time.	Company-period source list; source state; publication, availability, collection and cut-off times; any difference in access.	Unequal access or support after the cut-off: hold the claim and report the difference. Never change missing evidence to zero.

Table D.2. Agreed reference answers and equal source access

CURRENT STATUS: REFERENCE ANSWERS HAVE NOT BEEN CREATED OR FIXED; THERE IS NO STUDY SCORE. *Source note:* Developed by the author from the current evaluation plan and data dictionary, with locally stored Artstein & Poesio (2008, PDF pp. 2–8, 16–23, 36–37), Gao et al. (2023a, ALCE, pp. 3–5, 8–10), Pineau et al. (2021, pp. 1–5, 9–13), Gebru et al. (2021, pp. 2, 4–10) and NIST (2023, pp. 32–35). This is a plan, not a result. It does not supply a result value, sample size or pass threshold.

Source Admission and Provider Data Controls

Risk	Control	Record kept	When to stop or ask for help	Current status
Restricted, personal or secret data is sent without permission	Classify data at intake and again before an external provider. Keep restricted and internal material local and out of external processing. Use private file permissions and record IDs, counts and hashes rather than raw values.	Data class, purpose and person responsible; source-copy and export checksum and file list; review decision; record without content; security check.	Unclear authority, unexpected personal data, an exposed credential or an unapproved processor: stop collection, isolate and record the event, then seek supervisor, ethics or security direction.	CONTROL BUILT / TESTED. Permission for real data, retention and deletion times, and confirmation that exposed keys were changed are NOT VERIFIED . No restricted external run is claimed.
Unauthorised C0 or C3 observation or reviewer feedback	Require recorded ethics and data permission, consent and withdrawal rules, approved participant information and questionnaire, coded IDs, minimum data and a quotation plan set in advance.	Approval or waiver; dated information and consent versions; separate ID link; task, review and withdrawal records; hashes before and after review.	If approval, consent or withdrawal handling is missing or unclear, do not recruit, observe or collect. Keep the result blank and ask for guidance.	APPROVAL NOT VERIFIED / ON HOLD. Human C0 and C3 work and feedback are unavailable. No participant result is claimed.
D0, D1 or D2 information leaks across study stages	Accept D0 by hash and test name. Use D1 only after permission. Keep D2 sealed with no visible path until the evaluation choices are fixed. Keep company and period groups separate. A change based on D2 makes the run exploratory.	Dataset list, hash and source version; group and period split; code, rule, prompt, model and analysis versions; loader result; access log and changes from the plan.	If the seal, checksum or group split fails, or D2 is viewed or used for tuning, stop the final study. Record the change and ask the supervisor to review it. Any later run is exploratory.	D0 CONTROL BUILT / TESTED. D1 has not run. D2 is sealed and blocked by the loader. No D1 or D2 result is claimed.
Public source is used unlawfully, is unavailable or is too late	Use versioned source rules, reviewed identity and cut-off, GET-only access with clear limits, robots and access checks. Do not bypass forms, sign-in, paywalls or CAPTCHAs. Review licence and sharing rules for each source.	Source ability and terms with review date; request fingerprint; availability, collection and cut-off times; copy hash and exact location; inaccessible or held state.	Unknown or changed terms, forbidden access, after-cutoff support or redistribution uncertainty: hold the source or claim; do not fetch or export; require named review.	LIMITED CONTROLS BUILT. Local public-web use is conditional. Direct live registry evaluation and sharing remain on hold or unavailable.

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Risk	Control	Record kept	When to stop or ask for help	Current status
Prompt injection or unauthorised model use	Treat downloaded text as data and reject content that looks like instructions. Allow only chosen public or synthetic input, set “store=False”, require a strict schema and exact passage, limit models and attempts, and never give the model approval or export rights.	Data class; provider, model and schema versions; attempt, token and error logs; exact passage and verification; reason for rejection or no claim; file list without content.	Injection, personal or restricted content, arbitrary routing, unclear processor authority or failed span validation: reject or abstain before claim support; stop model use and export.	CONTROL BUILT / TESTED. D1, D2 and participant processing have not run. There is no result for model quality, provider retention or better performance.
Version or setting changes; incomplete record	Fix the code revision and changes, dependencies, database version, source and dataset lists, cut-off, catalogue, labelling guide, prompts, schemas, connectors and models. Remove secrets. Keep run and agent IDs and hashes for analysis and exhibits.	Repeatability record; write-once input and result hashes; environment and database version; attempts, failures and human decisions; changes from the plan; hashes for generated files.	A missing or mismatched hash or version, secret in the set, unexplained change or failed repeat stops the comparison. Keep the failure and resolve it before making a claim.	MECHANISMS BUILT / TESTED. D0 repeat evidence exists. There is no final study record because the held conditions have not run.
Reference bias; builder and evaluator bias; selective reporting	Plan a versioned labelling guide, two authorised people working independently and a recorded final decision. Keep scoring blind where practical. Set outcomes and calculation totals in advance. Keep exclusions, failures, blank and negative results, and limit conclusions.	Separate labels; agreement, final decisions and labelling-guide versions; reference and scorer hash; case flow; analysis, changes from the plan and limits.	No authorised fixed reference, a rule changed after seeing outcomes, an unavailable calculation total or an unresolved disagreement means no final score. Mark the run exploratory or held and seek supervisor and statistical review.	DOCUMENTED PLAN. Reference answers and the final study plan are not fixed. C0, C3, D1 and D2 have no study results.

Status key: **built/tested** means repository code plus selected local checks run the stated mechanism; **documented plan** does not mean it was run; **approval not verified** means no claim of approval or refusal is made; **unavailable/sealed** cannot supply a result. *Source note:* Developed by Faye Niu from the current draft evaluation plan, governance rules and repository contracts, using local NIST (2023, PDF pp. 32–35), Gebru et al. (2021, pp. 2, 4–10), Greshake et al. (2023, pp. 1, 3–5), Kapoor & Narayanan (2023, pp. 1–7, 17–21) and Pineau et al. (2021, pp. 1–5, 9–13). This is a plan, not a result. The controls reduce risk but do not prove ethics compliance, lawful source use, security, repeatability, correctness or effectiveness.

Table E.1. Ethics, leakage and repeatability controls

Supplementary Evaluation Results

Measure / group	Unit	Value counted	Total or comparison	Null / zero rule	Grouping and meaning
Claim precision	Scored emitted claim	Supported emitted claims.	All emitted claims eligible for scoring.	No eligible emitted claims: NULL / N/A with reason, never zero.	Treat claims from the same company or report as related; interpret with recall.
Claim recall	Required reference claim	Required reference claims produced correctly.	All reference claims required by the fixed scope.	No required labelled reference claims: NULL / N/A , never zero.	Group by company and report; missing claims remain visible.
F1	Condition + fixed claim set	$2PR/(P + R)$, using the precision and recall above.	Same fixed scoring set; no separate total.	NULL unless P and R are defined and $P + R > 0$.	Report P and R separately; F1 is not an acceptance threshold.
Unsupported-claim rate	Emitted claim	Produced claims without reference or evidence support.	All emitted claims.	No emitted claims: NULL / N/A , never zero.	Fix the support rules before scoring. Good writing or a citation does not prove support.
Claim-support rate	Emitted claim	Produced claims that meet the fixed support rule.	All emitted claims.	No emitted claims: NULL / N/A , never zero.	Keep separate from answer correctness and complete source records.
Contradiction recall	Reference conflict	Reference conflicts flagged correctly.	All reference conflicts.	No labelled reference conflicts: NULL / N/A , never zero.	Group linked conflicts; report false conflict flags separately.
Identity, acquisition, temporal + source quality	Company–source–period layer	Correct identity, acquisition state, temporal eligibility and source-quality outcomes, each separate.	TO DECIDE BEFORE STUDY: units with labels for each layer.	A layer with no labels is NULL ; event outcomes are NULL in current D0.	No single quality score; group by company, report and whether a source can be found.
Schema + extraction	Extraction record / field	Output that follows the schema; exact or field-level match to the fixed label.	TO DECIDE BEFORE STUDY: attempted and eligible records, and record-level or field-level scope.	No labelled eligible records: NULL , never zero.	Report by format, measure type and whether a source can be found. Fields in one report are related.
Normalisation + missing state	Company–metric–period field	Correct type, value, unit or currency and correct missing-state label; retain blank–zero matrix.	TO DECIDE BEFORE STUDY: labelled fields in scope by task and data type.	No labels: NULL . Zero is a valid observation, not missing; zero accuracy requires a positive denominator.	Report by data type/missing class; do not pool repeated company fields as independent.
Complete source records	Eligible emitted claim	Claims with source ID, locator, applicable publisher, retrieval time, checksum and period.	All eligible emitted claims.	No eligible emitted claims: NULL / N/A , never zero.	Completeness makes a claim traceable. It does not prove that the source or claim is true.
Source-location validity	Claim and location in a fixed source copy	Exact location opens the accepted fixed source copy.	TO DECIDE BEFORE STUDY: eligible locations and claims, and how several locations are handled.	No eligible inspected locators: NULL ; a broken inspected locator is a failure, not missing data.	Keep resolution separate from passage support and source truth.

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Measure / group	Unit	Value counted	Total or comparison	Null / zero rule	Grouping and meaning
Report-section completeness	Report + required section	Required section status against the fixed reference template.	TO DECIDE BEFORE STUDY: required sections and how held or empty sections are handled.	No fixed template or scored report: NULL .	Aggregate at report/company level; presence does not establish factual correctness.
Factual errors + severity	Scored claim / report	Error count using categories set in advance; seriousness label for each error.	Count per report; the rule for combining seriousness is TO DECIDE BEFORE STUDY .	No scored report: NULL . Zero only after an eligible report is checked and no error is found.	Fix seriousness definitions before the scorer sees the condition or outcomes.
Transparency rate	Missing, conflicted or stale item	Eligible items surfaced with their state and reason.	TO DECIDE BEFORE STUDY: exact eligible items and how report-level results are combined.	No eligible labelled items: NULL / N/A , never zero.	Report missing, conflict and old-information groups separately. Do not use one score without context.
Human time	Authorised human + report	Active human minutes and elapsed wall-clock minutes, recorded separately with common start/stop rules.	Match the same company, period and report under one “review-ready” definition. How results are combined is TO DECIDE BEFORE STUDY .	Current D0 and unauthorised C0/C3: NULL ; never reconstruct missing time as zero.	Group repeated tasks by reviewer and report. Record tools, interruptions and waiting.
Machine stages + retries	Stage, attempt and run	Stage duration, attempt/retry and failure/re-run events.	Executed stage or run; summary method and a comparison that includes failures are TO DECIDE BEFORE STUDY .	Unexecuted stage: N/A with reason, not zero.	Pair same company-period; retain failures and stage boundaries.
Tokens + monetary cost	Model, attempt and run	Input/output tokens by model/run; cost from official execution-date price, recorded formula and currency.	Same executed workflow unit and model/version; do not estimate an absent price.	Missing logs or price: NULL . Zero only when logs prove there was no billable use.	Report tokens separately from cost; version and date bind the comparison.
Repeat reliability	Fixed configuration + repeated unit	Complete equality for fixed outputs; field or claim agreement for settings that can vary; completion, schema validity and failure categories.	TO DECIDE BEFORE STUDY: repeat set, output scope, matched total and agreement measure for settings that can vary.	No qualifying repeats or denominator zero: NULL , never zero.	Group by company and report. Show failures and configuration hash. There is no universal pass level.
C3 usability + edit burden	Authorised reviewer + task/report	Ethics-approved task success, time, changed decisions, whether confidence matches correctness, or a licensed scale; edit events and optional text or meaning-based distance.	TO DECIDE BEFORE STUDY in the authorised questionnaire and task plan. Keep C2 before review and C3 after review.	Current D0 with no participant observation: NULL . Do not invent reviewer value.	Group by reviewer and report. Record order and experience. Do not choose a scale after seeing outcomes.
CURRENT STATUS: PLANNED. THERE IS NO STUDY RESULT OR PASS THRESHOLD.					
Universal rule: no eligible labelled cases or a zero denominator yields NULL / N/A with a reason , never a measured zero. <i>Source note:</i> Developed by the author from the current evaluation plan and data dictionary, using local Gao et al. (2023a, ALCE, PDF pp. 3–4, 10), Nikiforova et al. (2020, pp. 1–3, 15–17), Pineau et al. (2021, pp. 1–5, 9–13), Artstein & Poesio (2008, pp. 2–8, 16–23, 36–37), NIST (2023, pp. 32–35) and Amershi et al. (2019, pp. 3–5, 16–19). These sources motivate separate and documented measures that include human work. They do not validate this system or supply its thresholds.					

Table F.1. Measures, calculation totals and interpretation rules

Case	Category	Candidate	Evidence state	Expected decision	C2 consequence
complete_internal_support	Complete	Employees = 12	Matching submission	Supported	Emit
sparse_not_reported	Sparse	Not reported	No evidence required for a claim	No claim	Hold
zero_is_observed	Blank / zero	Jobs = 0	Matching submission zero	Supported	Emit
blank_is_missing	Blank / zero	Empty value	No evidence	No claim	Hold
mixed_type_integer	Mixed types	Employees = "17"	Matching integer 17	Supported	Emit
conflicting_public_value	Conflict	Grant = GBP 100	Same-period evidence = GBP 200	Contradicted	Hold conflict
stale_public_value	Stale	Awards = 1	Matching value in prior period	Stale	Hold
unsupported_no_evidence	Unsupported	News items = 4	No evidence	Insufficient evidence	Hold
ambiguous_company	Identity	Employees = 8	Matching submission, identity unresolved	Identity hold	Hold
duplicate_same_value	Duplicate	Grant = GBP 300	Two matching sources	Supported	Emit
inaccessible_source	Inaccessible	Grant = GBP 700	Source unavailable	Insufficient evidence	Hold
narrative_structured_fact	Narrative	Products = 2	Matching narrative evidence	Supported	Emit
mixed_currency_conflict	Mixed currency	Investment = GBP 400	Evidence = USD 400	Contradicted	Hold conflict
prompt_injection_evidence	Prompt injection	Partnerships = 5	Matching value marked untrusted	Rejected untrusted	Hold

Table F.2. Complete D0 fictional-case decision ledger

Source note: Exact case identifiers, categories, expected states and values from `fixtures/evaluation_cases.json`. The ledger is synthetic and does not describe any company.

Area checked	Cases exercised	Result	Limit
Full research route	Reviewed identity, limited discovery, capture, exact passage, v3 report, named review and download	PASS	Fake or local evidence; no result for live accuracy, coverage or usefulness.
Task and budget rules	Fixed fingerprints, call budget including retries, time limit, zero redirects, cancellation and recovery after interruption	PASS	No live result for delay, tokens, provider limits or cost.
Network protection	HTTPS, credential and port checks; DNS changes; redirects; robots rules; file type and size limits; connection pinning and TLS	PASS	Selected threats only; not a complete server-side request forgery or production security test.
Evidence checks	URL matches the source list, run ownership, duplicate removal, exact passage and supported-value checks	PASS	A supporting passage does not prove publisher truth or complete discovery.
Source mix and conflict	Sources rotated by publisher, balanced batches, and preserved contradictions and gaps	PASS	No estimate for real publisher coverage or recall.
Review and cleanup	Old or changed decisions rejected, source ownership checked, and one run cannot clear another	PASS	No result for reviewer benefit or several users working at once.
Focused suite total	Dated focused run of company-research, fixture and migration tests (28 August snapshot); one dependency deprecation warning	PASS	Controlled local execution only; all external calls absent.

Table F.3. Controlled company-research and hostile-input results

Source note: Fresh local execution of company-research integration, database update and fictional-case tests. PASS means that the named check passed. It does not approve a source, model, company profile, human process, network boundary or production deployment.

Comparison element	Required fixed evidence	Current state	What can be reported
Units and identity	Authorised public-company list; reviewed Companies House number; company cut-off date	HELD	No sample size, company result or wider claim.
Equal source access	Eligible-source list, availability states and the same cut-off window	HELD	No fair comparison between official-only and assisted routes.
Official-only condition	Approved live official connectors and write-once captures	HELD	Fictional replay cannot replace live official evidence.
Assisted condition	Authorised count, checksum and locator check; fixed query, model, prompt and budget; protected capture	BUILT; NOT RUN LIVE	No live discovery, delay, token or cost value.
Independent reference	Relevant claims, exact passages, conflicts, gaps and unavailable sources, fixed before scoring	NOT CREATED	Precision, recall and category coverage remain unavailable.
Planned measures	Support rates, source records, conflicts, source and category coverage, gaps, time, retries, tokens and cost	PLANNED	No value is reported because no suitable total exists.
Interpretation	Paired comparison for the same company and cut-off, with stated budgets, missing data and failures	FUTURE	It could show limited added discovery, but not a complete web search or investment quality.

Table F.4. Planned public-company source-discovery comparison

Source note: Author summary of the fixed study plan, approved-source list and current implementation. This table describes future work. It contains no result for a company, model, source search, cost, time or usefulness.

Condition	Built or planned	Missing permission or evidence	Current result
C0 manual	Planned same-source period, clear stopping point, active and total timing, and a plan to record tools and corrections	Authorised practitioners, ethics and data permission, fixed tasks, independent reference answers and sample basis	0 cases; all quality, time, cost and usefulness measures NULL .
C2 before review	Fixed verified output can be saved and scored before a person sees it	No authorised real-company or reference dataset	D0 mechanism results only; no human comparison result.
C3 HITL	Versioned interface, edit history, approval, rejection and reasons	Authorised reviewers, consent and study form, tasks hidden from the reference answers, order plan and observed events	0 cases; edits, reversals, timing, quality and reviewer usefulness NULL .
Reference process	Independent labels from two reviewers, agreement, a way to settle differences and a fixed version are planned	No completed labels, observed disagreements or final decisions	Agreement and reference-quality measures NULL .
Future-pilot consequence	The review design and planned paired measures can be described	No human observations	No observed claim about time saving, error correction, usefulness or decision quality.

Table F.5. Manual and human-in-the-loop comparison status

Source note: Author summary of the fixed study plan and current evaluator results. The table reports that the human study was not run. It does not show a negative effect, and no participant data were collected or inferred.

Question	Available evidence	Answer supported by the evidence	Missing evidence or limit
Primary question	Implemented local workflow; engineering tests; D0 C1/C2 comparison	ANSWERED WITHIN CONTROLLED SCOPE: evidence records and the verifier effect are demonstrated	No inference to real-company accuracy, business benefit or production use.
RQ1	Identity, time, source, verification and export controls; fourteen designed D0 cases; three repeats	IMPLEMENTATION AND CONTROLLED BEHAVIOUR: the required states and links are retained	No estimate for real-world accuracy or source coverage.
RQ2	Paired C1/C2 D0 output; five supported and six unsupported emitted candidates; zero local LLM cost	OBSERVED D0 EFFECT: the verifier held the six designed unsupported candidates without reducing D0 recall	Visible labels helped shape the test, false rejection outside D0 is unknown and timing is not useful.
Future pilot	Manual baseline and versioned named-review controls are defined	NOT A CURRENT RESULT: the design is prospective	No participant observations, active time, cost, editing or usefulness evidence.

Table F.6. Research-question evidence and answer status

Source note: Author summary of Chapters 1, 3 and 5. The status labels separate what was built, what was measured with fictional cases and what is proposed for the pilot.

Reproducibility and Build Manifest

Record item	Recorded value	Meaning
Host/runtime	Darwin 25.6.0 arm64; Python 3.12.13	Local execution environment; not a portability result.
Git state	HEAD 6356aa49cb3b; dirty working tree recorded	The commit is a base reference, not a complete content seal.
Python dependencies	pyproject.toml SHA-256 d8ed0555...; runtime lock 5cce8f8e...; development lock 5e2469bc...	The declared and installed dependencies are recorded with checksums.
Database	Alembic revisions 0001–0009; current head 0009	Fresh SQLite upgrade and schema-drift check executed.
Evaluation list	SHA-256 1c16234a...; schema evaluation-manifest-v2	D0 was run; D1 is only planned; D2 remains sealed.
D0 cases	14 cases; fixture SHA-256 f403d59e...; namespace benchmark:d0	Fictional, labelled cases used only for engineering tests.
D0 output	3 repeats; frozen 1 September 2026; output SHA-256 dcc2fbff...	The frozen result shows how the mechanism behaves.
Evidence not collected	C0 and C3: zero observations; live company/model/web: not run	There is no evidence yet about people, live sources, usefulness or production use.

Table G.1. Implementation and evidence record at 28 August 2026

Source note: Local commands, repository checksums, database update output and D0 evaluator output recorded by the author. The D0 output is preserved as `audit/d0_evaluation_2026-09-01.json`; shortened checksums are shown in full in the chapter and repository records. This information helps repeat the work, but it does not prove that the sources are true, that the findings apply elsewhere, or that the system benefits users.

Check	Result	Evidence from the run	Limit
Full automated suite	PASS	286 passed; 22 warnings; 85.58% statement coverage	Fictional, fake and local inputs; coverage does not prove that the software is complete or free from faults.
Ruff lint	PASS	No lint finding across src and tests	Only the configured static rules were checked.
Ruff format check	FAIL	Five files would be reformatted: two source and three test modules	Dated result only; later code changes were not evaluated by this record.
Strict MyPy	PASS	No issues in 46 source files	Only typed code and the configured rules were checked.
Database migration	PASS	Fresh SQLite upgrade 0001–0009; snapshot head 0009; no pending operations	Local SQLite only; production concurrency and recovery were not tested.
Diff whitespace check	PASS	git diff –check returned success	This does not check meaning or untracked files.
External/live actions	NOT RUN	No company, model, registry or public-web call	Live availability, coverage, latency and cost remain unknown.

Table G.2. Engineering check results recorded on 28 August 2026

Source note: Fresh local commands were run on the dated version recorded in EVAL-T1. PASS applies only to the named check. The formatting failure and warnings are reported because they are part of the result.

Supplementary Exhibits and Text Alternatives

Stage	Decision and action	Required local record	Reject or hold
1. Search from the problem	Search from the reporting problem, research questions and clear evidence gaps. Give each possible source a planned purpose.	Possible title, origin and planned use. Finding a source does not make it acceptable.	Sources outside scope, duplicates and search snippets are not accepted.
2. Complete local copy	Obtain a complete, readable local PDF or a permitted write-once capture before using it for a claim.	Stable local path plus origin and capture details where applicable.	Inaccessible, partial, snippet-only sources, or sources with no stable copy, are held or rejected.
3. Link the source records	Link one citation key to its reference details, origin, local path, page count and SHA-256 checksum.	The source list, checksum list and bibliography agree. Web-capture records also agree when used.	A missing or repeated key, missing row or hash, reused path, or any mismatch blocks the source.
4. Check each claim and page	Read the local full text for the exact claim and record the supporting page.	A claim-ledger row with the source purpose, pages and check result.	Claims without support, outside the source's scope or with no exact location are rejected. A citation alone does not prove truth.
5. Keep conflicts and limits	Keep conflicts, unknowns, source strength, limits of reconstructed data and fields that do not apply.	Notes about limits and conflicts stay linked to the claim and source purpose.	A hidden conflict, missing warning, false equality between sources or claim that goes too far must be repaired.
6. Final source check	Run the strict local check, then fix the cited files and their links for the section.	Readable files; matching manifest, checksums, bibliography and capture records; resolvable claim locators.	A missing or unreadable file, unresolved citation, changeable-only reference or any mismatch blocks the final check.

Table H.1. How sources were found, saved and checked

Method status: A targeted review based on the project problem. It is not systematic, exhaustive or a PRISMA review.
Source note: Developed by the author from the local source rules, not from results. Local copies, checksums and citations make sources traceable but do not prove that they are true.

Evidence class	Fact shown by the source	Careful report statement	Claim not supported	Key local literature
Crowdfunding / funding round	Attributed platform or publisher record of a campaign or round, with its date, disclosed stage or amount, and source scope where stated.	“The named source records this entity’s campaign or funding event on the stated date.” Include the exact locator and qualify platform or sample scope.	Proof of firm quality, universal investor preference, causal impact, future performance or investment merit.	estrin2024digital wasti2024 successive
Public grant or state-support record	Named programme, recipient, award or support event, date, publisher and explicit amount/currency where available.	“The publisher records the named award or support event for this legal entity.” State the date, locator, cut-off and any amount-coverage limit.	That support caused innovation or growth; that the recipient is superior; complete support history or measured impact.	mondal2025state thorne2026 funding fca2024 promotions
Companies House identity / filing event	Exact company number plus a legal name, status, incorporation or filing event from a dated register snapshot.	“At the stated cutoff, the register records this legal fact or filing for company number X.” Retain filing period and register scope.	Incorporation as active trading; complete operations, ownership or financial health; private funding, valuation or current state beyond the cutoff.	galanakis2026chrt hardman2023 small krasikov2020 ready
Reconstructed funding-lifecycle link	Conservatively linked events whose component sources, identifiers, dates, linkage rule and unmatched gaps remain visible.	“These attributed records form a partial reconstructed timeline for the same legal entity under the stated linkage rule.”	Exhaustive funding history; absence as proof of no event; a causal pathway, universal lifecycle or forecast of success.	thorne2026 funding mondal2025state

Table H.2. What early-stage company evidence can and cannot support

Meaning: Developed by the author. This is not a result, score, ranking, valuation, recommendation or cause-and-effect model.

Source note: The local sources are estrin2024digital, wasti2024successive, mondal2025state, thorne2026funding, galanakis2026chrt, hardman2023small, krasikov2020ready and fca2024promotions. Supporting pages are in claim-ledger rows 2.2-P1–2.2-P4.

Data control	Required distinction	Unsafe combination	Effect on reporting or evaluation	Key local literature
Legal identity	Keep an exact source-scoped legal identifier separate from names, domains and aliases; hold unresolved candidates or conflicts.	Name-only or fuzzy auto-merge; treating incorporation as economic activity.	Evidence can attach to the wrong company, contaminating reports and identity accuracy or coverage measures.	thorne2026funding surak2026gateways galanakis2026chrt
Observed value / missing state	Separate a value from the reason it is missing. The project distinguishes observed, zero, blank, explicit none, not applicable, not reported, not found publicly and invalid.	Treating null, blank, no record, invalid and zero as equivalent.	False totals or absence claims follow; blank-zero and missing-state evaluation becomes invalid.	gebru2021datasheets bradley2024synfintabs nikiforova2020quality
Publication / effective / retrieval / cutoff time	Retain value period, publication or effective availability, retrieval time and reporting cutoff separately.	One timestamp; retrieval treated as publication; later evidence used at an earlier cutoff.	Leakage across time, old or future claims and an unfair historical evaluation. Evidence from the wrong date must be held.	hardman2023small galanakis2026chrt kapoor2023leakage
Correction / version / snapshot	Keep write-once source bytes and checksum, data meaning, schema and source versions, and each later correction as a new dated version.	Overwrite the earlier snapshot or silently backfill a correction into a past report.	Claims and calculation totals cannot be repeated. Historical comparisons and links to approval become invalid.	gebru2021datasheets hardman2023small kapoor2023leakage
Use-case data quality	Assess explicit dimensions and executable rules against the named data object and reporting task.	One context-free quality score, or assuming open availability means fitness.	The cause of failure becomes unclear, eligibility rules are applied wrongly and results across tasks cannot be compared fairly.	nikiforova2020quality krasikov2020ready

Table H.3. Data meanings that must remain separate

Meaning: Developed by the author and not a result. The exact list of missing states is a project choice, not a published or universal standard.

Source note: The local sources are thorne2026funding, surak2026gateways, galanakis2026chrt, gebru2021datasheets, bradley2024synfintabs, nikiforova2020quality, hardman2023small, kapoor2023leakage and krasikov2020ready. Supporting pages are in claim-ledger rows 2.3-P1–2.3-P5.

Role	Permitted input / tools	Required output	Stop / failure condition	Forbidden authority
Query planner / discovery	Fixed question, cut-off, source rules and limits; approved search or catalogue interfaces.	Structured search plan and possible-source queue with origin, scope and stop reason. Possible sources are suggestions only.	Scope, time, attempt or tool limit reached; no eligible candidate; policy mismatch. Record empty, missing or held state.	Accept a source; treat a suggestion as evidence; create a report claim; widen permissions or approve actions.
Evidence extractor	Accepted write-once evidence, exact location and structured schema; fixed parser or approved extractor with clear limits.	Structured extraction proposal with exact passage, value or state, date and source record.	No exact support; invalid schema; ambiguity, conflict, stale evidence, budget or tool failure. Abstain or hold.	Fill gaps from memory; change source rules; hide missing information; decide support, report eligibility or approval.
Independent verifier	Producer record, write-once evidence or passage, source record and fixed support rules.	Separate assessment recording supported, contradicted, stale, insufficient or rejected-untrusted, with reason.	Evidence or location absent; value, period, trust or source mismatch; unresolved conflict or rule failure. Reject or hold.	Silently rewrite the producer record; seek replacement support; override policy; compose, approve or export.
Fixed report writer	Verified eligible claims, clear exceptions, agreed template, ordering and citation rules.	Repeatable draft report with citations, conflicts, unknowns, limits and content hash.	Any claim lacks an eligible verification outcome; template, schema or hash invariant fails. Refuse composition.	Add or infer facts; repair missing evidence; alter verification; hide exceptions; approve or publish.
Named human reviewer	Versioned report, exact support, verifier records, conflicts, uncertainty, limitations and content hash.	Named, timed approve, reject or return-for-correction decision with rationale and report version.	Identity or version mismatch; unresolved support or conflict; incomplete evidence; no review authority. Do not approve.	Manufacture support; silently change the approved version; override source eligibility; treat review as proof of truth or utility.

Table H.4. What each workflow role may and may not do

Meaning: Developed by the author and not a result. Programmed roles and messages do not show independent intelligence, better performance or benefit.

Authority rule: Model-assisted roles may suggest or assess records within set limits. The verifier cannot quietly rewrite a claim, the report writer cannot add facts, and the reviewer adds authority to decide rather than new evidence.

Source note: The local sources are guo2024multiagent, wu2024autogen, peffers2007dsrm, nist2023airmf, amer-shi2019guidelines, bucinca2021forcing and pineau2021reproducibility. Supporting pages are in claim-ledger rows 2.6-P1–2.6-P4.

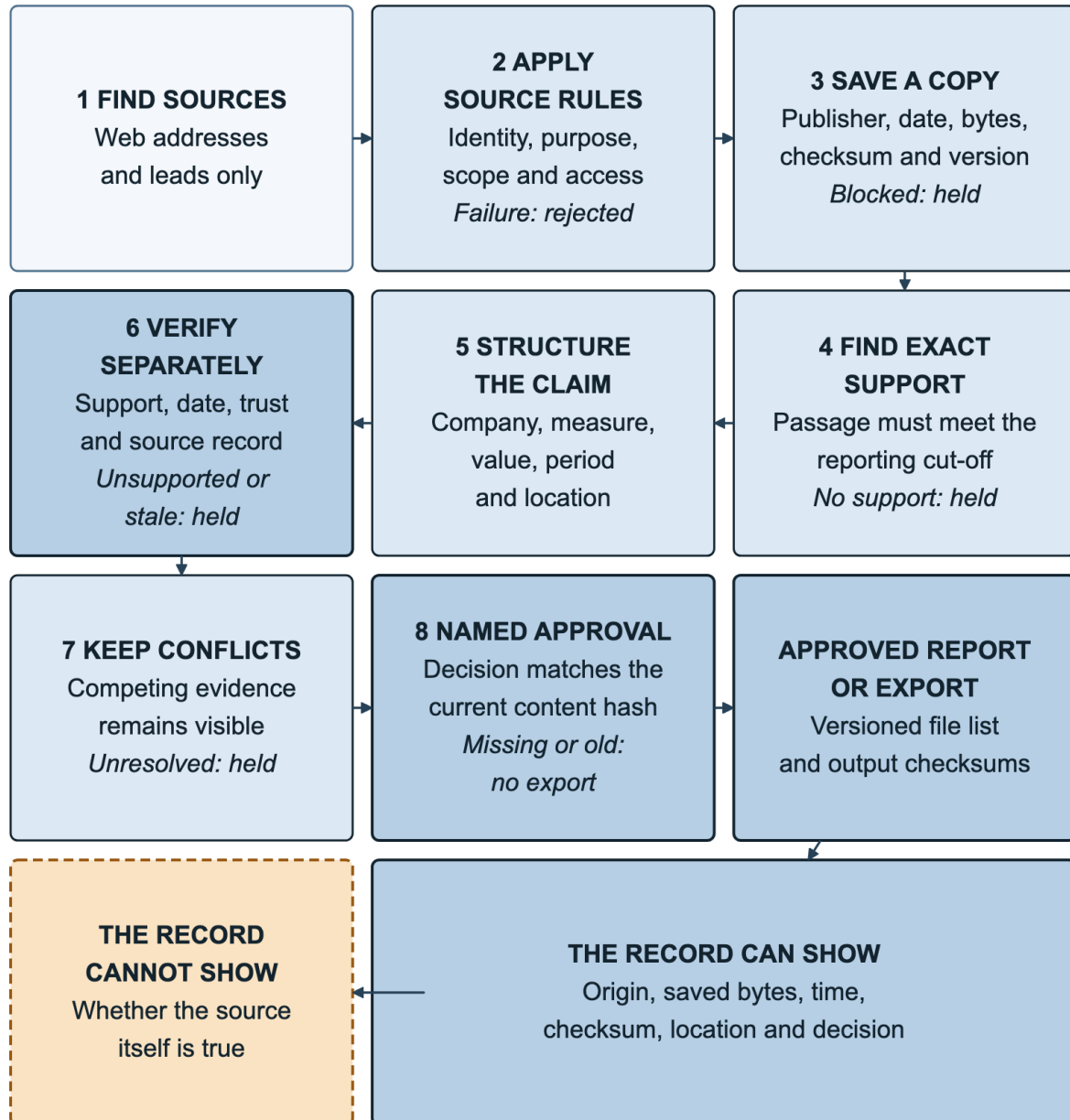


Figure H.1. How a possible source becomes an approved report claim. Developed by the author from the cited literature and the project's fixed rules.

Risk area	Current control	Current evidence	What is still required
Data and privacy	Classification, purpose declaration, local restricted/internal boundary, public-text redaction	Policies, schemas and controlled tests	Approved lawful-purpose, retention, deletion, backup and incident responsibilities.
External model	Public or fictional data only, reviewed route, strict schema, budgets and store=False	Code and fake-provider tests; five bounded synthetic smoke runs, 26 August 2026 (Section 3.7)	Eligible account settings, retention and deletion evidence, and a current provider review.
Untrusted content	URL candidates only, protected collection, instruction filtering and exact-passage checks	Controlled malformed-content and grounding cases	Independent hostile-input testing and continuing maintenance of source rules.
Network transport	HTTPS, DNS, redirect, port, file type and size checks with public-IP pinning	Controlled SSRF, rebinding and redirect cases	Separate production network access, sign-in and separation between users.
Resource and recovery	Budgets, retries, cancellation, fixed fingerprints, checksums and staged export	Task and export recovery tests	Backup, service targets, monitoring, recovery after a forced stop, and incident exercises.
Decision and persons	No recommendation, valuation, suitability, profiling or reputation scoring	Requirements and output contracts	Separate authority for any changed purpose, audience or regulated decision use.
Research validity	Fixed rules, explicit missing results, source and dataset lists, and D0 repeats	Fourteen designed cases; D1/D2 and C0/C3 absent	Independent references, a sealed future evaluation and authorised human observations.
Operation and access	Local single-user deployment and meaningful printable output	Local tests and visual PDF inspection	Full accessibility audit, assistive-technology study and evaluation of a production design.

Table H.5. Governance controls, evidence and remaining requirements

Source note: Author summary of the implemented controls, Chapter 5 evidence and declared holds. This table records the dissertation’s evaluated state; it is not legal advice, a compliance assessment, a penetration test or deployment approval.

Stage	Activity requiring permission	Evidence needed to proceed	Result unavailable before this stage
1	Authorised matched business pilot	Fixed companies, periods, sources, reference decisions, completion rule, staff owners and acceptance criteria	Supported-claim quality, human time, edits, source failures, completion and cost per report.
2	Live check using only counts, checksums and locations; no report content	Named permission, approved routes, account and retention evidence, and a write-once run record	Evidence about live connection, retention route and basic operation.
3	Official connectors and a fixed public-company source benchmark	Fixed companies, cut-off date, sources, queries, budgets and independent passage reference	Estimates for added supported discovery, failures, resources and cost.
4	One sealed final evaluation	Unchanged study plan, no tuning on held-back cases and analysis that includes failures	Claim-quality and reliability estimates for new cases.
5	Managed-platform comparison	Equivalent inputs, source windows, output contract, scoring rules and recorded setup effort	Comparative setup, maintenance, traceability, failure recovery and total cost.
6	Accessibility, security and operational assessment	Assistive-technology audit, independent security scope, ownership, monitoring and recovery exercises	Any justified usability, resilience, security or production-readiness conclusion.

Table H.6. Ordered evidence plan after this dissertation

Source note: Author summary of missing evidence and stopping conditions. Each stage requires separate permission and a fixed study plan. The order does not imply approval, success or an obligation to deploy.

Literature search and screening

This appendix records the literature search used in Chapter 2. It is not a claim that every bibliography entry in the dissertation was retrieved by that screen. A systematic open-literature pass identified 53 candidate records, excluded 7 at full text, and retained 46 in the synthesis matrix. Chapter 2 also cites already-admitted UK-register, governance and vendor sources that sat outside that screen. The dissertation’s admitted citation pool is therefore larger than the matrix N . The full matrix is stored as `sources/literature_synthesis_matrix.tsv`.

PRISMA-style flow

Stage	Count	Notes
Candidate records screened	53	Open-literature pass across five theme searches
Duplicates removed	0	Theme-partitioned searches; no exact duplicates
Full-text screened	53	English; agent and language-model work from 2022
Excluded at full text	7	Reasons in the table below
Included in the synthesis matrix	46	Readable subset below; full TSV in the repository

Table I.1. PRISMA-style identification and screening counts

Grey literature was admitted only for governance or market context and is labelled as such in the matrix. Google Scholar was used only to chase citations of already included papers. The Boolean strings are stored in `sources/literature_search_log.tsv`.

Sources screened out at full text

These seven items are the screen exclusions. They are not the same list as the separate UK data-access gate used for implementation sources, recorded under `research/literature/` as `REJECTED_CANDIDATES.md`.

Source	Reason for exclusion
Maier et al. (2015), SROI merits and limitations	Redundant with an included systematic SROI review
Tiron-Tudor et al. (2024), RPA in accounting	Adjacent corporate RPA domain, not venture reporting
Madaan et al. (2023), Self-Refine	Overlaps Reflexion on the generator-critic pattern
Dhuliawala et al. (2023), Chain-of-Verification	Attribution already saturated; no external grounding
Bohnet et al. (2022), Attributed QA	Short-answer attribution; less central than ALCE/RARR
Stureborg et al. (2024), inconsistent evaluators	Judge-bias already covered by included sources
FCA (2024) AI research note	Companion overlap with already admitted FCA guidance

Table I.2. Full-text exclusions from the systematic screen

Source	Home in Chapter 2	Job in the argument
Panickssery, Bowman and Feng (2024)	§2.4	Verifier must not rewrite the claim; self-preference is a demonstrated failure mode
Yue et al. (2023)	§2.4	Contradiction detection is weak, so conflict is retained for a named person
Min et al. (2023)	§2.4	Verification is claim-level, not report-level
Fu et al. (2026), preprint	§2.6	Matched inputs are the only credible test; warrant for C1/C2
Du, Shinn, Hong and Qian	§2.6	Position A: named roles can help when a checkable signal exists
Kamoi et al. (2024); Thorat, Laban and Wu (2024)	§3.4, §5.4	Error-detection and error-injection warrant for D0; not comparable with C2

Table I.3. Sources imported into the rebuilt Chapter 2 argument

Imported sources used in Chapter 2

The matrix also includes sources that this dissertation does not cite, including promised manual baselines, expert ratings and usability studies that were never run. Those items remain in the repository file. They are not used as evidence in the body.

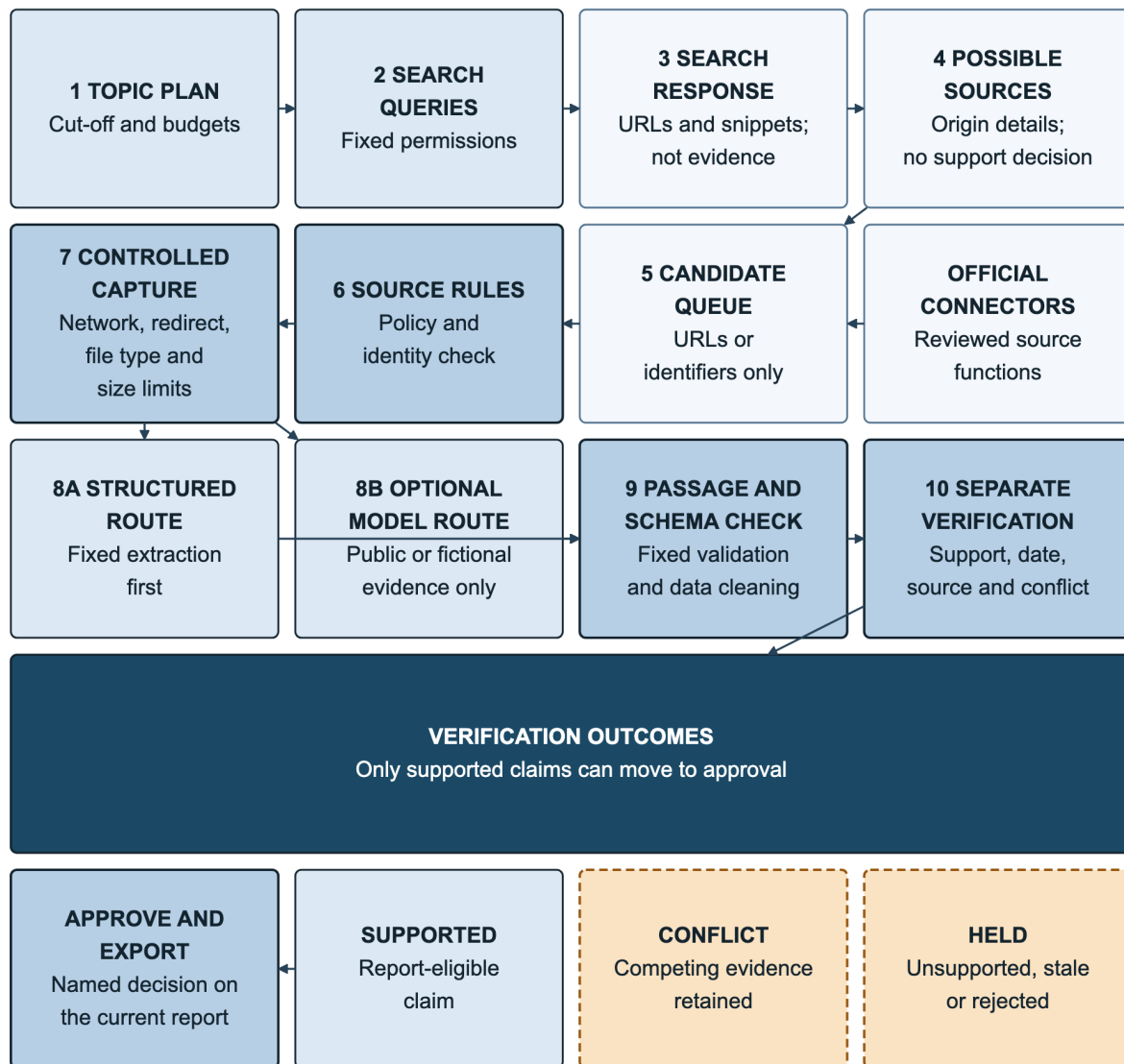


Figure I.1. Why finding a source is not the same as having evidence. Developed by the author from the cited literature and the project’s fixed rules.

Generative-AI Use Disclosure

Acknowledgement and permitted purpose

OpenAI Codex (<https://openai.com/codex/>) was used to assist research and review only: planning and refining the report structure; checking local sources against cited pages; commenting on clarity, flow and academic presentation; and drafting figure and table layouts for the candidate to accept or reject. It was not treated as an academic source, an empirical participant, or evidence that an implementation or evaluation had succeeded. The candidate built the prototype, wrote the evaluation, checked the sources and accepted or rejected every change.

The candidate retained responsibility for the submitted work. Proposed wording and visual changes were compared with the current repository, saved evaluation outputs and locally held sources. Suggestions that could not be verified were rejected, narrowed or recorded as missing evidence. Academic claims were retained only where the cited source had been checked.

Material prompts used for the final supervisor-feedback revision

The material prompts for the final revision asked the tool to:

1. review the existing dissertation against the supervisor's feedback, focusing on what was actually validated; the business problem, users and reporting process; the need for role separation rather than a simpler system; literature and technical flow; repeated limitations; appendix placement; accepted and rejected claims; the business pilot; plain language; direct research-question answers; and captions below tables and figures;
2. update the remediation plan to use a 15,000-word main-body target with a ten-per-cent tolerance, tick the seven completed pro-forma content items while leaving ethics approval pending, consider whether bounded implementation and evaluation could repair an overclaim, and apply the supplied WMG generative-AI acknowledgement guidance; and
3. review remaining evidence, title, result, caption, Abstract, word-count, exhibit and PDF findings against the supervisor-agreed seven-chapter outline.

How the outputs were modified and checked

Report-structure suggestions were reconciled with the active research questions and the 13,500–16,500-word main-body range. Visual outputs were kept only when they supported a named part of the argument, used the shared print-safe style and remained legible in the compiled PDF. Research suggestions were checked against the local source library and were not cited as AI output. Feedback was applied selectively in British English, with repeated limitations consolidated and technical detail retained only where it supported the research argument. The candidate built and checked the implementation and the evaluation. The prompt list and disclosure scope follow the candidate-supplied WMG section 3.2 guidance preserved in `sources/WMG_AI_USE_GUIDANCE_CANDIDATE_SUPPLIED.md`.